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(Accredited by NBA for Civil, EEE, Mech., ECE & CSE

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Puttur -517583, Tirupati District, A.P. (India)

Department of Computer Science and Engineering

(Common to CSE, CSM & CAD)

(20CS0519) OBJECT ORIENTED ANALYSIS AND DESIGN LAB

III B.Tech - I Semester

Lab Observation Book

Academic Year: _____

Name : _____

Roll. Number : _____

Year & Branch: _____

Specialization : _____

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(Accredited by NBA for Civil, EEE, Mech., ECE & CSE
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Puttur -517583, Tirupati District, A.P. (India)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INSTITUTE VISION

To emerge as one of the premier institutions through excellence in education and research, producing globally competent and ethically strong professionals and entrepreneurs

INSTITUTE MISSION

M1: Imparting high-quality technical and management education through the state-of-the- art resources.

M2: Creating an eco-system to conduct independent and collaborative research for the betterment of the society

M3: Promoting entrepreneurial skills and inculcating ethics for the socio-economic development of the nation.

DEPARTMENT VISION

To impart quality education and research in Computer Science and Engineering for producing technically competent and ethically strong IT professionals with contemporary knowledge

DEPARTMENT MISSION

M1: Achieving academic excellence in computer science through effective pedagogy, modern curriculum and state-of-art computing facilities.

M2: Encouraging innovative research in Computer Science and Engineering by collaborating with Industry and Premier Institutions to serve the nation.

M3: Empowering the students by inculcating professional behavior, strong ethical values and leadership abilities

SIDDHARTH INSTITUTE OF ENGINEERING & TECHNOLOGY

(AUTONOMOUS)
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(Accredited by NBA for Civil, EEE, Mech., ECE & CSE
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Puttur -517583, Tirupati District, A.P. (India)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Program Outcomes

PO1: Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct Investigations of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1: To provide software solutions for arising problems in diverse areas with strong knowledge in innovative technologies of computer science.

PEO2: To serve in IT industry as professionals and entrepreneurs or in pursuit of higher education and research.

PEO3: To attain professional etiquette, soft skills, leadership, ethical values meld with a commitment for lifelong learning.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: **Analysis & Design:** Ability to design, develop and deploy customized applications in all applicable domains using various algorithms and programming languages.

PSO2: **Computational Logic:** Ability to visualize and configure computational need in terms of hardware and software to provide solutions for various complex applications.

PSO3: **Software Development:** Ability to apply standard procedures, tools and strategies for software development.

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LAB INCHARGE

SIDDARTHA INSTITUTE OF ENGINEERING AND TECHNOLOGY :: PUTTUR (AUTONOMOUS)

III B.Tech. – I Sem.

L	T	P	C
-	-	3	1.5

(18CS0520) OBJECT ORIENTED ANALYSIS AND DESIGN LAB

COURSE OBJECTIVES

The Objectives of this Course:

1. *Illustrate the requirements specification for an intended software system*
2. *Demonstrate the UML diagrams for the given specification*
3. *Map the design properly to the code*

COURSE OUTCOMES (COs)

On successful completion of the course students will be able to:

1. *Construct various UML models (including use case diagrams, class diagrams, interaction diagrams, state chart diagrams, activity diagrams, and implementation diagrams) using the appropriate notation*
2. *Describe the importance of systems analysis and design in solving complex problemsExplain how the object-oriented approach differs from the traditional approach to systems analysis and design*
3. *Understand the role and function of each UML model in developing object oriented software*
4. *Exhibit software development process*
5. *Recognize the difference between various object relationships: inheritance, association, whole-part, and dependency relationships*

List of Experiments:

Draw standard UML diagrams using an UML modeling tool for a given case study and map design to code and implement a 3 layered architecture. Test the developed code and validate whether the SRS is satisfied.

1. Identify a software system that needs to be developed.
2. Document the Software Requirements Specification (SRS) for the identified system.
3. Identify use cases and develop the Use Case model.
4. Identify the conceptual classes and develop a Domain Model and also derive a Class Diagram from that.
5. Using the identified scenarios, find the interaction between objects and represent them using UML Sequence and Collaboration Diagrams
6. Draw relevant State Chart and Activity Diagrams for the same system.

7. Implement the system as per the detailed design
8. Test the software system for all the scenarios identified as per the use case diagram
9. Improve the reusability and maintainability of the software system by applying Appropriate design patterns.
10. Implement the modified system and test it for various scenarios

OOAD Problems that may be considered are

1. College information system
2. Hostel management
3. ATM system
4. Library management system
5. Passport Automation System
6. Political Administration System.

TEXT BOOKS:

1. Grady Booch, James Rumbaugh, Ivar Jacobson, *The Unified Modeling Language, User Guide,* 2nd edition, Pearson Education, 2005.
2. Hans-Erik Eriksson, Magnus Penker, Brian Lyons, David Fado, *UML 2 Toolkit,* WILEY- Dreamtech India Pvt. Ltd, 2009.

Ex. No.: 1	Case Tool
Date:	

INTRODUCTION TO UML

Contents:

1. Importance of Modeling
2. Principles of Modeling
3. Object Oriented Modeling
4. Conceptual Model of the UML
5. Architecture
6. Software Development Life Cycle

1. Importance of Modeling:

Why do we model?

A model is a simplification at some level of abstraction

We build models to better understand the systems we are developing.

- To help us visualize
- To specify structure or behavior
- To provide template for building system
- To document decisions we have made

2. Principles of Modeling:

- The models we choose have a profound influence on the solution we provide
- Every model may be expressed at different levels of abstraction
- The best models are connected to reality
- No single model is sufficient, a set of models is needed to solve any nontrivial system

UML is a visual modeling language

"A picture is worth a thousand words." - old saying

Unified Modeling Language:

"A language provides a vocabulary and the rules for combining words [...] for the purpose of
communication.
A modeling language is a language whose vocabulary and rules focus on the conceptual and physical representation of a system. A modeling language such as the UML is thus a standard language for software blueprints."

Usages of UML:

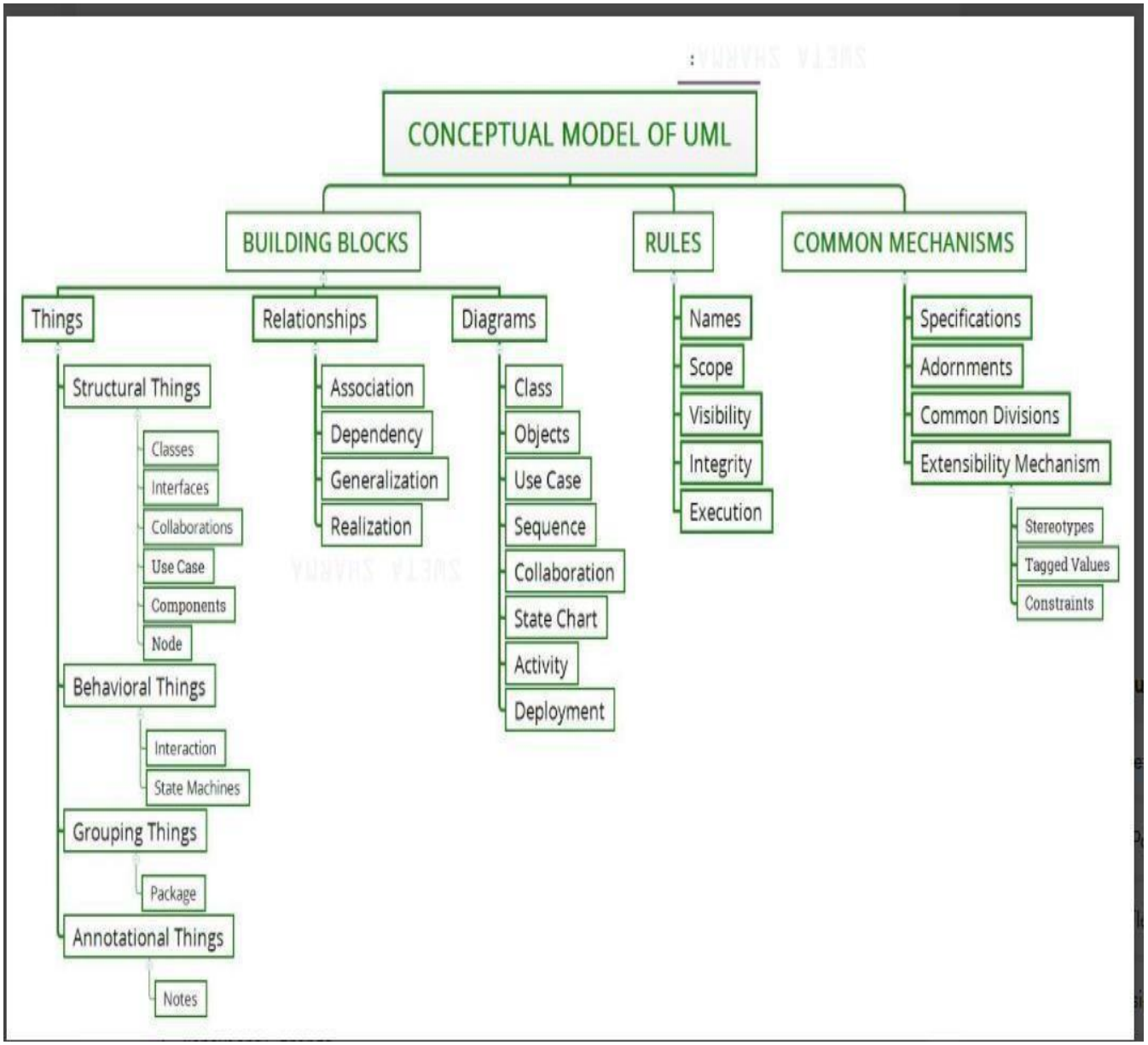
UML is used in the course to

- document designs patterns / frameworks
- represent different views/aspects of design – visualize and construct designs static / dynamic / deployment / modular aspects
- provide a next-to-precise, common, language –specify visually for the benefit of analysis, discussion, comprehension...

Conceptual Model of the UML:

To understand the UML, you need to form a conceptual model of the language, and this requires learning three major elements: the UML's basic building blocks, the rules that dictate how those building blocks may be put together, and some common mechanisms that apply throughout the UML. Once you have grasped these ideas, you will be able to read UML models and create some basic ones. As you gain more experience in applying the UML, you can build on this conceptual model, using more advanced features of the language.

Conceptual Model of UML



BUILDING BLOCKS OF THE UML

The vocabulary of the UML encompasses three kinds of building blocks:

1. Things
2. Relationships
3. Diagrams

Things are the abstractions that are first-class citizens in a model; relationships tie these things together; diagrams group interesting collections of things.

Things in the UML

There are four kinds of things in the UML:

1. Structural things
2. Behavioral things
3. Grouping things
4. Annotational things

These things are the basic object-oriented building blocks of the UML. You use them to write well-formed models.

Structural Things

Structural things are the nouns of UML models. These are the mostly static parts of a model, representing elements that are either conceptual or physical. Collectively, the structural things are called *classifiers*.

A ***class*** is a description of a set of objects that share the same attributes, operations, relationships, and semantics. A class implements one or more interfaces. Graphically, a class is rendered as a rectangle, usually including its name, attributes, and operations, as in Figure 2-1.

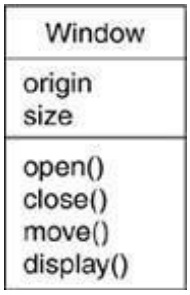


Figure 2-1. Classes:

An ***interface*** is a collection of operations that specify a service of a class or component. Aninterface therefore describes the externally visible behavior of that element. An interface might represent the complete behavior of a class or component or only a part of that behavior. An interface defines a set of operation specifications (that is, their signatures) but never a set of operation implementations. The declaration of an interface looks like a class with the keyword «interface» above the name; attributes are not relevant, except sometimes to show constants. An interface rarely stands alone, however. An interface provided by a class to the outside world is shown as a small circle attached to the class box by a line. An interface required by a class from some other class is shown as a small semicircle attached to the class box by a line, as in Figure 2-2.

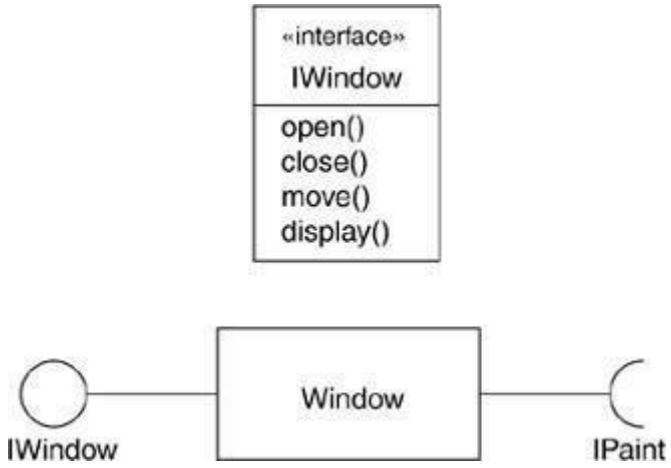


Figure 2-2. Interfaces

A *collaboration* defines an interaction and is a society of roles and other elements that work together to provide some cooperative behavior that's bigger than the sum of all the elements. Collaborations have structural, as well as behavioral, dimensions. A given class or object might participate in several collaborations. These collaborations therefore represent the implementation of patterns that make up a system. Graphically, a collaboration is rendered as an ellipse with dashed lines, sometimes including only its name, as in Figure 2-3.

Figure 2-2. The collaboration

A ***use case*** is a description of sequences of actions that a system performs that yield observable results of value to a particular actor. A use case is used to structure the behavioral things in model.

A use case is realized by a collaboration. Graphically, a use case is rendered as an ellipse with solid lines, usually including only its name, as in Figure 2-4.

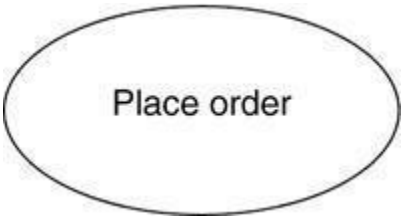


Figure 2-4. Use Cases

The remaining three things active classes, components, and nodes are all class-like, meaning they also describe sets of entities that share the same attributes, operations, relationships, and semantics. However, these three are different enough and are necessary for modeling certain aspects of an object-oriented system, so they warrant special treatment.

An *active class* is a class whose objects own one or more processes or threads and therefore can initiate control activity. An active class is just like a class except that its objects represent elements whose behavior is concurrent with other elements. Graphically, an active class is rendered as a class with double lines on the left and right; it usually includes its name, attributes, and operations, as in Figure 2-5.

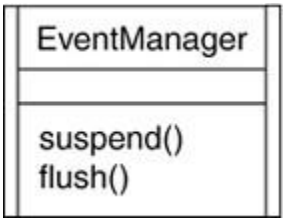


Figure 2-5. Active Classes.

component is a modular part of the system design that hides its implementation behind a set of external interfaces. Within a system, components sharing the same interfaces can be substituted while preserving the same logical behavior. The implementation of a component can be expressed by wiring together parts and connectors; the parts can include smaller components.

Graphically, a component is rendered like a class with a special icon in the upper right corner, as in Figure 2-6.

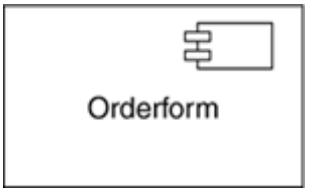


Figure 2-5. Component.

The remaining two elements artifacts and nodes are also different. They represent physical things, whereas the previous five things represent conceptual or logical things.

An *artifact* is a physical and replaceable part of a system that contains physical information ("bits"). In a system, you'll encounter different kinds of deployment artifacts, such as source code files, executables, and scripts. An artifact typically represents the physical packaging of source or run-time information. Graphically, an artifact is rendered as a rectangle with the keyword «artifact» above the name, as in Figure 2-7.

Figure 2-7. Artifacts:

A *node* is a physical element that exists at run time and represents a computational resource,generally having at least some memory and, often, processing capability. A set of components may reside on a node and may also migrate from node to node. Graphically, a node is rendered as a cube, usually including only its name, as in Figure 2-8.

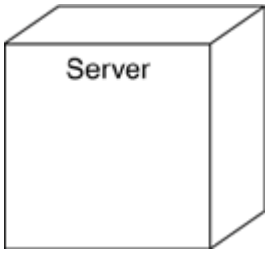


Figure 2-8. Nodes

These elements classes, interfaces, collaborations, use cases, active classes, components, artifacts, and nodes are the basic structural things that you may include in a UML model. There are also variations on these, such as actors, signals, and utilities (kinds of classes); processes and threads (kinds of active classes); and applications, documents, files, libraries, pages, and tables (kinds of artifacts).

Behavioral Things:

Behavioral things are the dynamic parts of UML models. These are the verbs of a model,representing behavior over time and space. In all, there are three primary kinds of behavioral things.

First, **an *interaction*** is a behavior that comprises a set of messages exchanged among a set of objects or roles within a particular context to accomplish a specific purpose. The behavior of a society of objects or of an individual operation may be specified with an interaction. An interaction involves a number of other elements, including messages, actions, and connectors (the connection between objects). Graphically, a message is rendered as a directed line, almost always including the name of its operation, as in Figure 2-9.

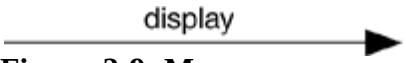


Figure 2-9. Messages

Second, **a *state machine*** is a behavior that specifies the sequences of states an object or an interaction goes through during its lifetime in response to events, together with its responses to those events. The behavior of an individual class or a collaboration of classes may be specified with a state machine. A state machine involves a number of other elements, including states, transitions (the flow from state to state), events (things that trigger a transition), and activities (the response to a transition). Graphically, a state is rendered as a rounded rectangle, usually including its name and its substates, if any, as in Figure 2-10.

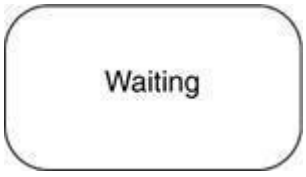


Figure 2-10. States

Third, **an activity** is a behavior that specifies the sequence of steps a computational process performs. In an interaction, the focus is on the set of objects that interact. In a state machine, the focus is on the life cycle of one object at a time. In an activity, the focus is on the flows among steps without regard to which object performs each step. A step of an activity is called an *action*.Graphically, an action is rendered as a rounded rectangle with a name indicating its purpose. States and actions are distinguished by their different contexts.

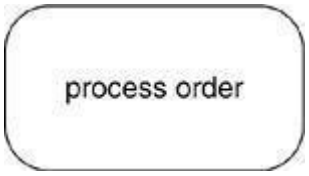


Figure 2-11. Actions

These three elements interactions, state machines, and activities are the basic behavioral things that you may include in a UML model. Semantically, these elements are usually connected to various structural elements, primarily classes, collaborations, and objects.

Grouping Things:

Grouping things are the organizational parts of UML models. These are the boxes into which a
model can be decomposed. There is one primary kind of grouping thing, namely, packages.
A *package* is a general-purpose mechanism for organizing the design itself, as opposed to classes, which organize implementation constructs. Structural things, behavioral things, and even other grouping things may be placed in a package. Unlike components

(which exist at run time), a package is purely conceptual (meaning that it exists only at development time). Graphically, a package is rendered as a tabbed folder, usually including only its name and, sometimes, its contents, as in Figure 2-12.

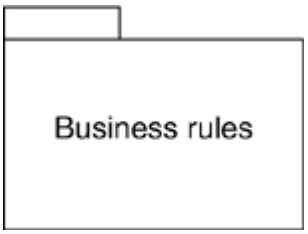


Figure 2-12. Packages

Packages are the basic grouping things with which you may organize a UML model. There are also variations, such as frameworks, models, and subsystems (kinds of packages).

Annotational Things:

Annotational things are the explanatory parts of UML models. These are the comments you may apply to describe, illuminate, and remark about any element in a model. There is one primary kind of annotational thing, called a note. A *note* is simply a symbol for rendering constraints and comments attached to an element or a collection of elements. Graphically, a note is rendered as a rectangle with a dog-eared corner, together with a textual or graphical comment, as in Figure 2-13.

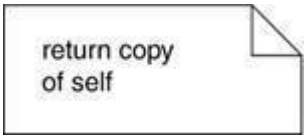


Figure 2-13. Notes

This element is the one basic annotational thing you may include in a UML model. You'll typically use notes to adorn your diagrams with constraints or comments that are best expressed in informal or formal text. There are also variations on this element, such as requirements (which specify some desired behavior from the perspective of outside the model).

RELATIONSHIPS IN THE UML

There are four kinds of relationships in the UML:

1. Dependency
2. Association
3. Generalization
4. Realization

These relationships are the basic relational building blocks of the UML. You use them to write well-formed models.

First, a *dependency* is a semantic relationship between two model elements in which a change to one element (the independent one) may affect the semantics of the other element (the dependent one). Graphically, a dependency is rendered as a dashed line, possibly directed, and occasionally including a label, as in Figure 2-14.

Figure 2-14. Dependencies

Second, an *association* is a structural relationship among classes that describes a set of links, a link being a connection among objects that are instances of the classes. Aggregation is a special kind of association, representing a structural relationship between a whole and its parts. Graphically, an association is rendered as a solid line, possibly directed, occasionally including a label, and often containing other adornments, such as multiplicity and end names, as in Figure 2-15.

Figure 2-15. Associations

Third, a *generalization* is a specialization/generalization relationship in which the specialized element (the child) builds on the specification of the generalized element (the parent). The child shares the structure and the behavior of the parent. Graphically, a generalization relationship is rendered as a solid line with a hollow arrowhead pointing to the parent, as in Figure 2-16.

Figure 2-16. Generalizations

Fourth, a *realization* is a semantic relationship between classifiers, wherein one classifier specifies a contract that another classifier guarantees to carry out. You'll encounter realization relationships in two places: between interfaces and the classes or components that realize them, and between use cases and the collaborations that realize

them. Graphically, a realization relationship is rendered as a cross between a generalization and a dependency relationship, as in Figure 2-17.

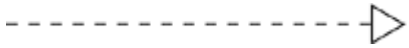


Figure 2-17. Realizations

These four elements are the basic relational things you may include in a UML model. There are also variations on these four, such as refinement, trace, include, and extend.

DIAGRAMS IN THE UML

A *diagram* is the graphical presentation of a set of elements, most often rendered as a connected graph of vertices (things) and paths (relationships). You draw diagrams to visualize a system from different perspectives, so a diagram is a projection into a system. For all but the most trivial systems, a diagram represents an elided view of the elements that make up a system. The same element may appear in all diagrams, only a few diagrams (the most common case), or in no diagrams at all (a very rare case). In theory, a diagram may contain any combination of things and relationships. In practice, however, a small number of common combinations arise, which are consistent with the five most useful views that comprise the architecture of a software-intensive system. For this reason, the UML includes thirteen kinds of diagrams:

1. Class diagram
2. Object diagram
3. Component diagram
4. Composite structure diagram
5. Use case diagram
6. Sequence diagram
7. Communication diagram
8. State diagram
9. Activity diagram
10. Deployment diagram
11. Package diagram
12. Timing diagram
13. Interaction overview diagram

A **class diagram** shows a set of classes, interfaces, and collaborations and their relationships. These diagrams are the most common diagram found in modeling object-oriented systems. Class diagrams address the static design view of a system. Class diagrams that include active classes address the static process view of a system. Component diagrams are variants of class diagrams.

An **object diagram** shows a set of objects and their relationships. Object diagrams represent static snapshots of instances of the things found in class diagrams. These diagrams address the static design view or static process view of a system as do class diagrams, but from the perspective of real or prototypical cases.

A **component diagram** is shows an encapsulated class and its interfaces, ports, and internal structure consisting of nested components and connectors. Component diagrams address the static design implementation view of a system. They are important for building large systems from smaller parts. (UML distinguishes a composite structure diagram, applicable to any class, from a component diagram, but we combine the discussion because the distinction between a component and a structured class is unnecessarily subtle.)

A **composite structure diagram** is a UML structural diagram that contains classes, interfaces, packages, and their relationships, and that provides a logical view of all, or part of a software system. It shows the internal structure (including parts and connectors) of a structured classifier or collaboration.

A **use case diagram** shows a set of use cases and actors (a special kind of class) and their relationships. Use case diagrams address the static use case view of a system. These diagrams are especially important in organizing and modeling the behaviors of a system.

Both **sequence diagrams** and **communication diagrams** are kinds of interaction diagrams.

An **interaction diagram** shows an interaction, consisting of a set of objects or roles, including the messages that may be dispatched among them. Interaction diagrams address the dynamic view of a system. A sequence diagram is an interaction diagram that emphasizes the time-ordering of messages.

A **communication diagram** is an interaction diagram that emphasizes the structural organization of the objects or roles that send and receive messages. Sequence diagrams and communication diagrams represent similar basic concepts, but each diagram emphasizes a different view of the concepts. Sequence diagrams emphasize temporal ordering, and communication diagrams emphasize the data structure through which messages flow. A timing diagram (not covered in this book) shows the actual times at which messages are exchanged.

A **state diagram** shows a state machine, consisting of states, transitions, events, and activities. These diagrams shows the dynamic view of an object. They are especially important in modeling the behavior of an interface, class, or collaboration and emphasize the event-ordered behavior of an object, which is especially useful in modeling reactive systems

An **activity diagram** shows the structure of a process or other computation as the flow of control and data from step to step within the computation. Activity diagrams address the dynamic view of a system. They are especially important in modeling the function of a system and emphasize the flow of control among objects.

Deployment diagram shows the configuration of run-time processing nodes and the components that live on them. Deployment diagrams address the static deployment view of an architecture. A node typically hosts one or more artifacts.

An **artifact diagram** shows the physical constituents of a system on the computer. Artifacts include files, databases, and similar physical collections of bits. Artifacts are often used in conjunction with deployment diagrams. Artifacts also show the classes and components that they implement. (UML treats artifact diagrams as a variety of deployment diagram, but we discuss them separately.)

A **package diagram** shows the decomposition of the model itself into organization units and their dependencies.

A **timing diagram** is an interaction diagram that shows actual times across different objects or roles, as opposed to just relative sequences of messages. An interaction overview diagram is a hybrid of an activity diagram and a sequence diagram. These diagrams have specialized uses and so are not discussed in this book. See the UML Reference Manual for more details.

This is not a closed list of diagrams. Tools may use the UML to provide other kinds of diagrams, although these are the most common ones that you will encounter in practice.

RULES OF THE UML

The UML's building blocks can't simply be thrown together in a random fashion. Like any language, the UML has a number of rules that specify what a well-formed model should look like. A well-formed model is one that is semantically self-consistent and in harmony with all its related models.

The UML has syntactic and semantic rules for

Name	What you can call things, relationships, and diagrams
Scope	The context that gives specific meaning to a name
Visibility	How those names can be seen and used by others
Integrity	How things properly and consistently relate to one another
Execution	What it means to run or simulate a dynamic model

Models built during the development of a software-intensive system tend to evolve and may be viewed by many stakeholders in different ways and at different times. For this reason, it is common for the development team to not only build models that are well-formed, but also to build models that are

Elided	Certain elements are hidden to simplify the view
Incomplete	Certain elements may be missing
Inconsistent	The integrity of the model is not guaranteed

These less-than-well-formed models are unavoidable as the details of a system unfold and churn during the software development life cycle. The rules of the UML encourage you but do not force you to address the most important analysis, design, and implementation questions that push such models to become well-formed over time.

COMMON MECHANISMS IN THE UML

A building is made simpler and more harmonious by the conformance to a pattern of common features. A house may be built in the Victorian or French country style largely by using certain architectural patterns that define those styles. The same is true of the UML. It is made simpler by the presence of four common mechanisms that apply consistently throughout the language.

1. Specifications
2. Adornments
3. Common divisions
4. Extensibility mechanisms

Specifications:

The UML is more than just a graphical language. Rather, behind every part of its graphical notation there is a specification that provides a textual element of the syntax and semantics of that building block. For example, behind a class icon is a specification that provides the full set of attributes, operations including their full signatures), and behaviors that the class embodies; visually, that class icon might only show a small part of this specification. Furthermore, there might be another view of that class that presents a completely different set of parts yet is still consistent with the class's underlying specification.

You use the UML's graphical notation to visualize a system; you use the UML's specification to state the system's details. Given this split, it's possible to build up a model incrementally by drawing diagrams and then adding semantics to the model's specifications, or directly by creating a specification, perhaps by reverse engineering an existing system, and then creating diagrams that are projections into those specifications.

The UML's specifications provide a semantic backplane that contains all the parts of all the models of a system, each part related to one another in a consistent fashion. The UML's diagrams are thus simply visual projections into that backplane, each diagram revealing a specific interesting aspect of the system.

Adornments:

Most elements in the UML have a unique and direct graphical notation that provides a visual representation of the most important aspects of the element. For example, the notation for a class is intentionally designed to be easy to draw, because classes are the most common element found in modeling object- oriented systems. The class notation also exposes the most important aspects of a class, namely its name, attributes, and operations.

A class's specification may include other details, such as whether it is abstract or the visibility of its attributes and operations. Many of these details can be rendered as graphical or textual adornments to the class's basic rectangular notation. For example, Figure 2-18 shows a class, adorned to indicate that it is an abstract class with two public, one protected, and one private operation.

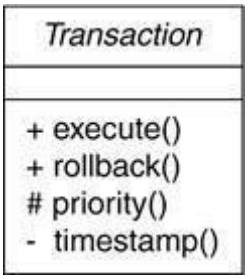


Figure 2-18. Adornments

Every element in the UML's notation starts with a basic symbol, to which can be added a variety of adornments specific to that symbol.

Common Divisions:

In modeling object-oriented systems, the world often gets divided in several ways.

First, there is the division of class and object. A class is an abstraction; an object is one concrete manifestation of that abstraction. In the UML, you can model classes as well as objects, as shown in Figure 2-19. Graphically, the UML distinguishes an object by using the same symbol as its class and then simply underlying the object's name.

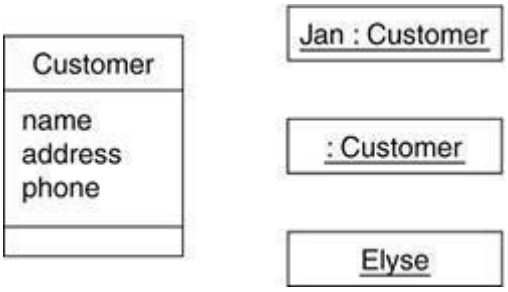


Figure 2-19. Classes and Objects

In this figure, there is one class, named Customer, together with three objects: Jan (which is marked explicitly as being a Customer object), :Customer (an anonymous Customer object), and Elyse (which in its specification is marked as being a kind of Customer object, although it's not shown explicitly here).Almost every building block in the UML has this same kind of class/object dichotomy. For example, you can have use cases and use case executions, components and component instances, nodes and node instances, and so on.

Second, there is the separation of interface and implementation. An interface declares a contract, and an implementation represents one concrete realization of that contract, responsible for faithfully carrying out the interface's complete semantics. In the UML, you can model both interfaces and their implementations, as shown in Figure 2-20.

Figure 2-20. Interfaces and Implementations

In this figure, there is one component named SpellingWizard.dll that provides (implements) two interfaces, IUnknown and ISpelling. It also requires an interface, IDictionary, that must be provided by another component.
Almost every building block in the UML has this same kind of interface/implementation dichotomy. For example, you can have use cases and the collaborations that realize them, as well as operations and the methods that implement them.

Third, there is the separation of type and role. The type declares the class of an entity, such as an object, an attribute, or a parameter. A role describes the meaning of an entity within its context, such as a class, component, or collaboration. Any entity that forms part of the structure of another entity, such as an attribute, has both characteristics: It derives some of its meaning from its inherent type and some of its meaning from its role within its context (Figure 2-21).

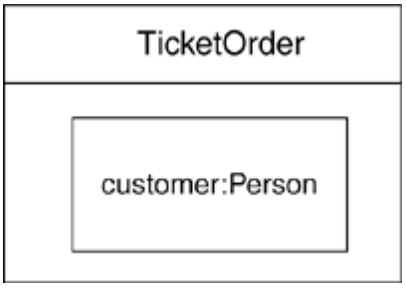


Figure 2-21. Part with role and type

Extensibility Mechanisms:

The UML provides a standard language for writing software blueprints, but it is not possible for one closed language to ever be sufficient to express all possible nuances of all models across all domains across all time. For this reason, the UML is opened-ended, making it possible for you to extend the language in controlled ways. The UML's extensibility mechanisms include .

- Stereotypes
- Tagged values
- Constraints

A *stereotype* extends the vocabulary of the UML, allowing you to create new kinds of building blocks that are derived from existing ones but that are specific to your problem. For example, if you are working in a programming language, such as Java or C++, you will often want to model exceptions. In these languages, exceptions are just classes, although they are treated in very special ways. Typically, you only want to allow them to be thrown and caught, nothing else. You can make exceptions first-class citizens in your dels meaning that they are treated like basic building blocks by marking them with an appropriate stereotype, as for the class Overflow in Figure 2-19.

A *tagged value* extends the properties of a UML stereotype, allowing you to create new information in the stereotype's specification. For example, if you are working on a shrink-wrapped product that undergoes many releases over time, you often want to track the version and author of certain critical abstractions. Version and author are not primitive UML concepts. They can be added to any building block, such as a class, by introducing new tagged values to that building block. In Figure 2-19, for example, the class EventQueue is extended by marking its version and author explicitly.

A *constraint* extends the semantics of a UML building block, allowing you to add new rules or modify existing ones. For example, you might want to constrain the EventQueue class so that all additions are done in order. As Figure 2-22 shows, you can add a constraint that explicitly marks these for the operation add.

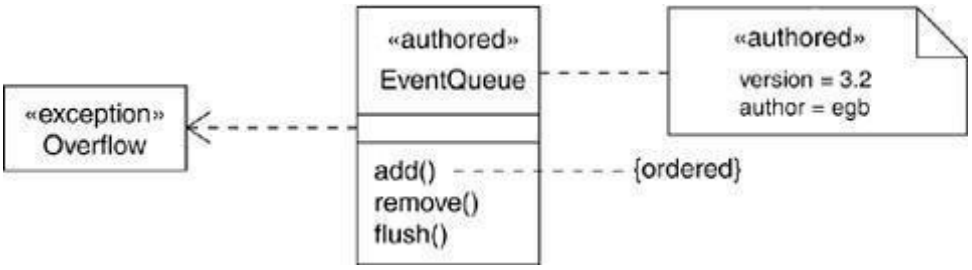


Figure 2-22. Extensibility Mechanisms

Collectively, these three extensibility mechanisms allow you to shape and grow the UML to your project's needs. These mechanisms also let the UML adapt to new software technology, such as the likely emergence of more powerful distributed programming languages. You can add new building blocks, modify the specification of existing ones, and even change their semantics.

Naturally, it's important that you do so in controlled ways so that through base extensions, you remain true to the UML's purpose the communication of information.

Ex. No.: 2	**College information system**
Date:	

AIM

To develop a project College Information System using Enterprise Architecture Software and to implement the software in Java.

PROBLEM ANALYSIS AND PROJECT PLANNING

A College Information System is a software application for educational establishments to manage College data. College Information System provide capabilities for entering student test and other assessment scores, building schedules, tracking student attendance, and managing many other College-related data needs in a school, college or College.

PROBLEM STATEMENT

a. Effective for Administration Purpose
b. Cheap
c. Better Service

UML DIAGRAMS:

The following UML diagrams describe the process involved in the online recruitment system

a. Use case diagram

b. Object diagram

b. Class diagram

c. Sequence diagram

d. Collaboration diagram

e. Activity diagram

f. Component diagram

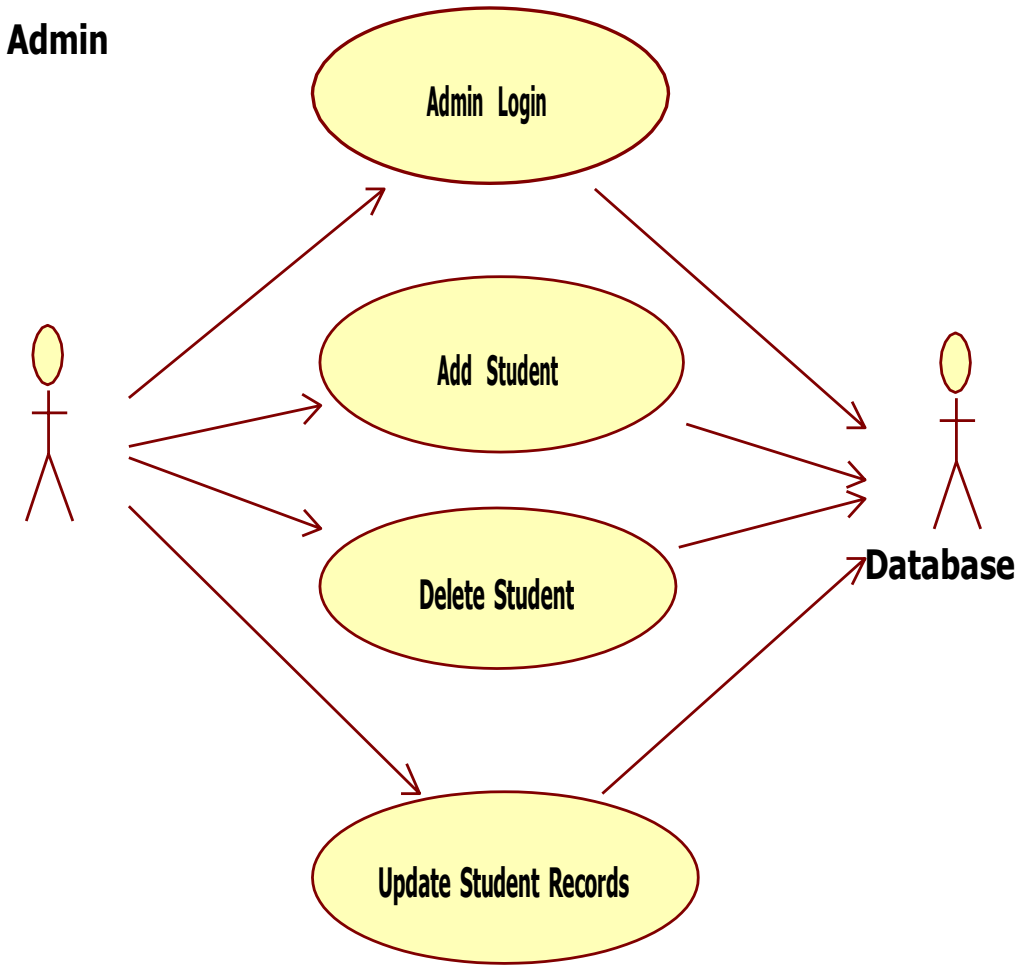
g. Data Flow Diagram

h. Deployment Diagram

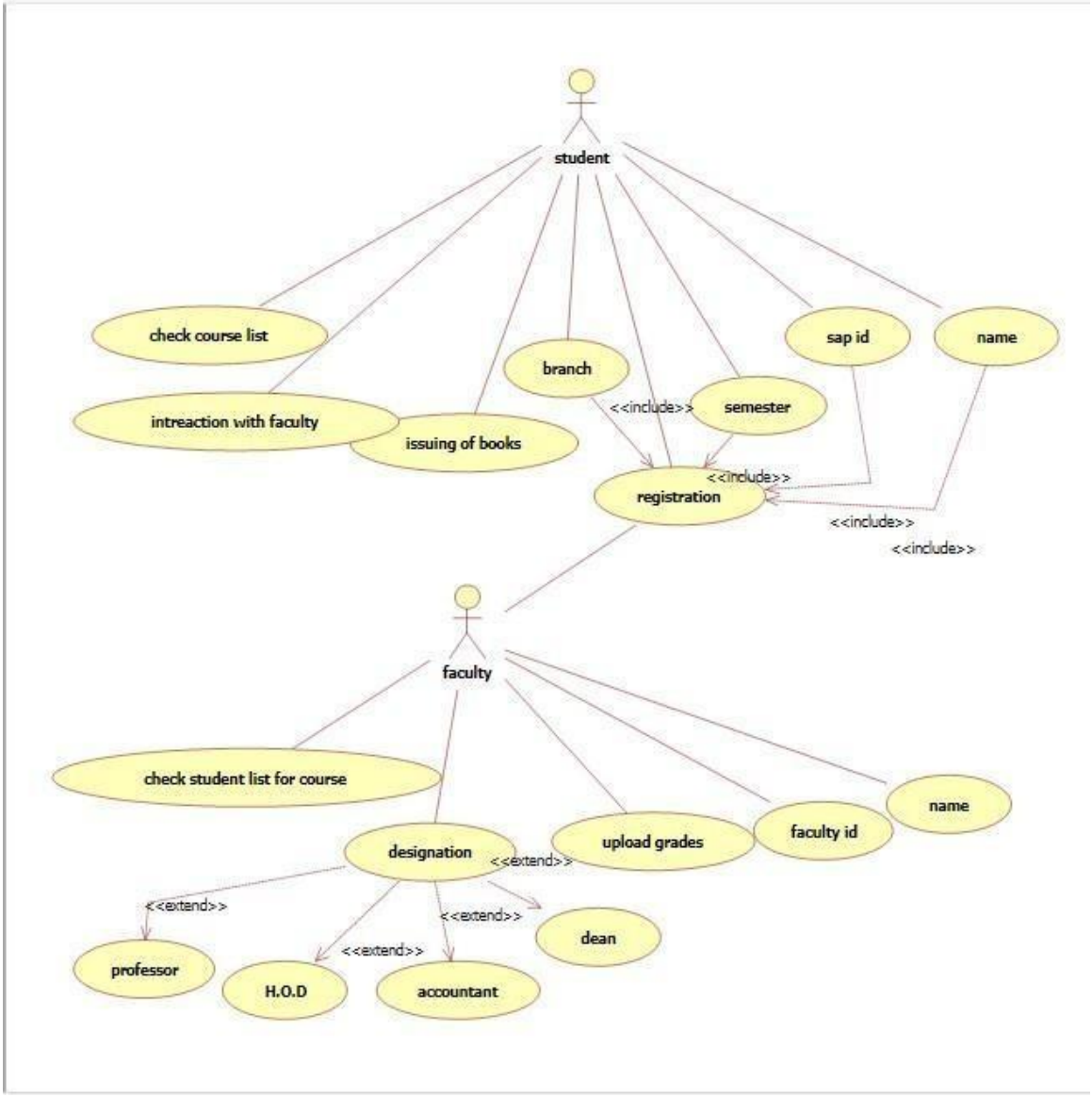
USE CASE DIAGRAM:

A use case is a methodology used in system analysis to identify, clarify, and organize system requirements. The use case is made up of a set of possible sequences of interactions between systems and users in a particular environment and related to a particular goal. It is represented using ellipse. Actor is any external entity that makes use of the system being modeled . It is represented using stick figure

For Administrator:



College



DOCUMENTATION OF USE CASE DIAGRAM

The actors in this use case diagram are Admin, Student, Database. The use cases are the activities performed by actors.

a. Admin register login, and store the student records details in database.

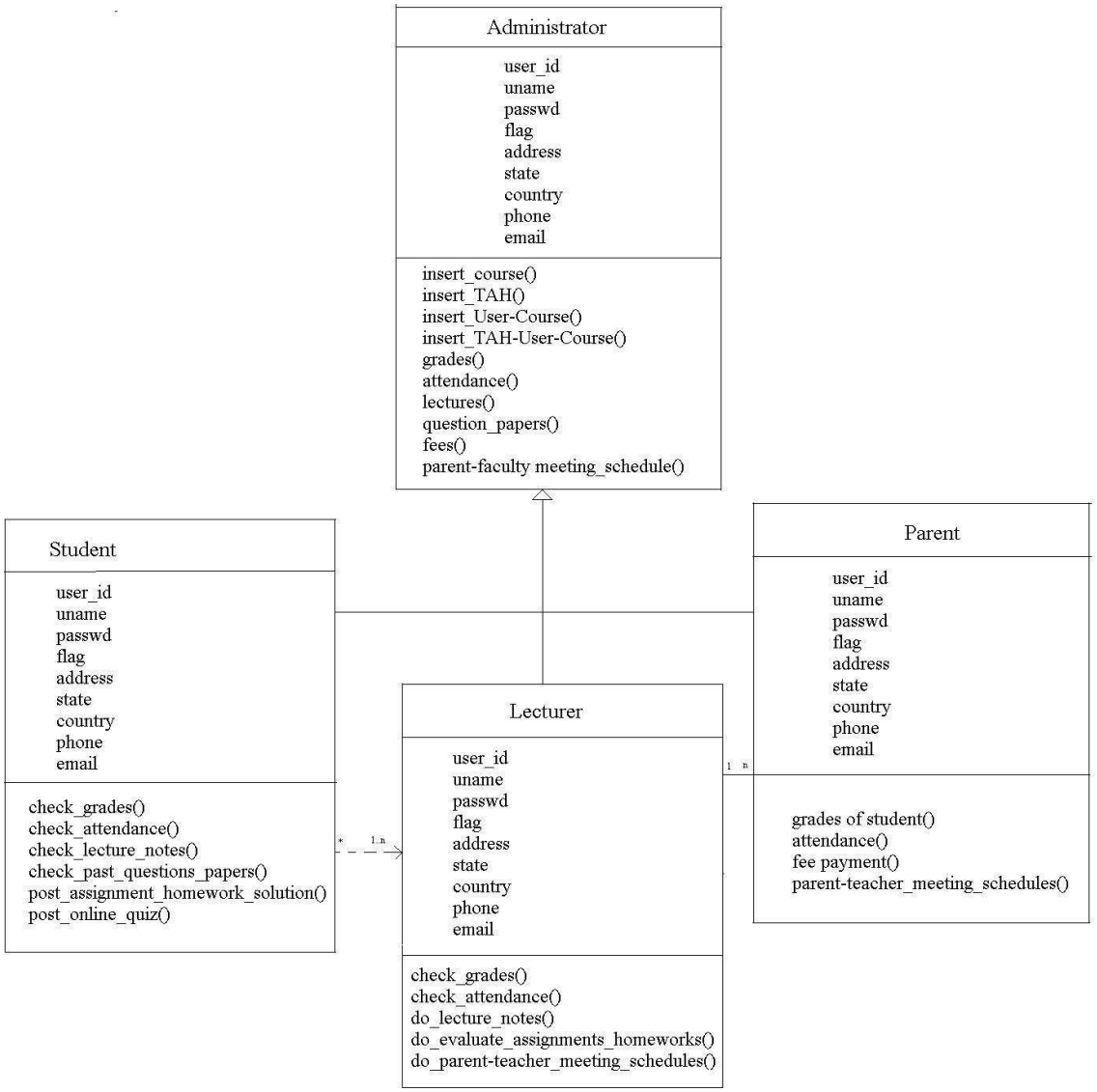
b. Student Register from the Student Login process.

c. Then the database is searched for details and verified.

d. Database stores the details and returns acknowledgement

CLASS DIAGRAM:

A class diagram in the unified modeling language (UML) is a type of static structure diagram that describes the structure of a system by showing the system's classes, their attributes, and the relationships between the classes. It is represented using a rectangle with three compartments. Top compartment have the class name, middle compartment the attributes and the bottom compartment with operations.



DOCUMENTATION OF CLASS DIAGRAM

This class diagram has three classes Login, Student details and Update details in database.

a. **Students** – is the class name. Its attributes are name, Address, DOB, Gender, College, Subjects, Semester, Year, Degree, Branch. The operations Performed in the students class, Store database and Update.

b. **Administration**– is the class name. Its attributes are Login, Password and database. The operations performed are Student Details store in database and send acknowledgement.

c. **Database** – is the class name. The operations performed are storing Search and storing the values.

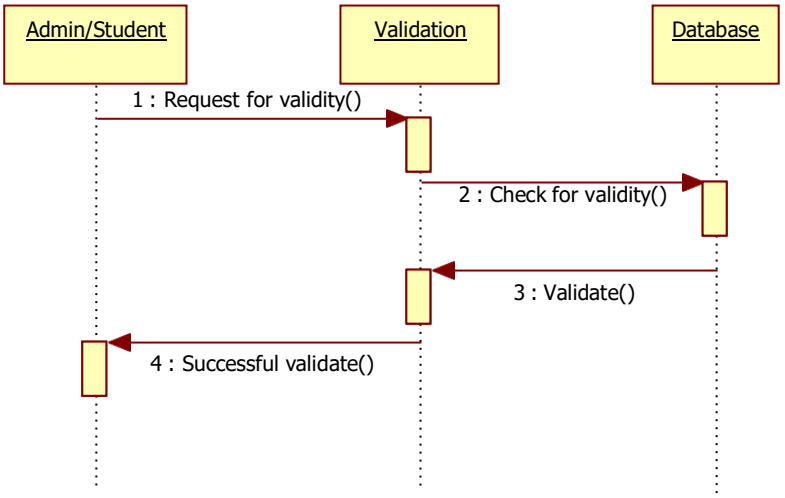
SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. There are two dimensions.

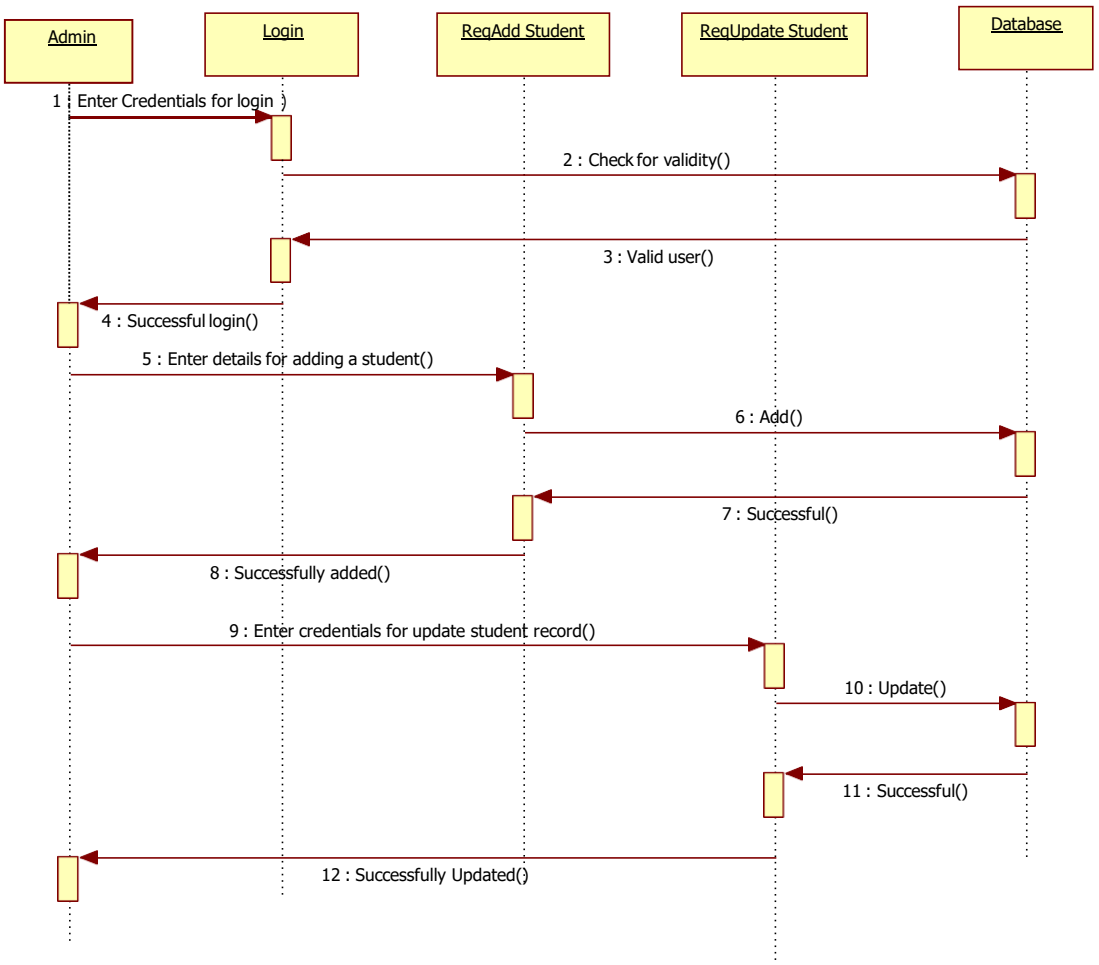
1. Veritcal dimension-represent time.

2. Horizontal dimension-represent different objects.

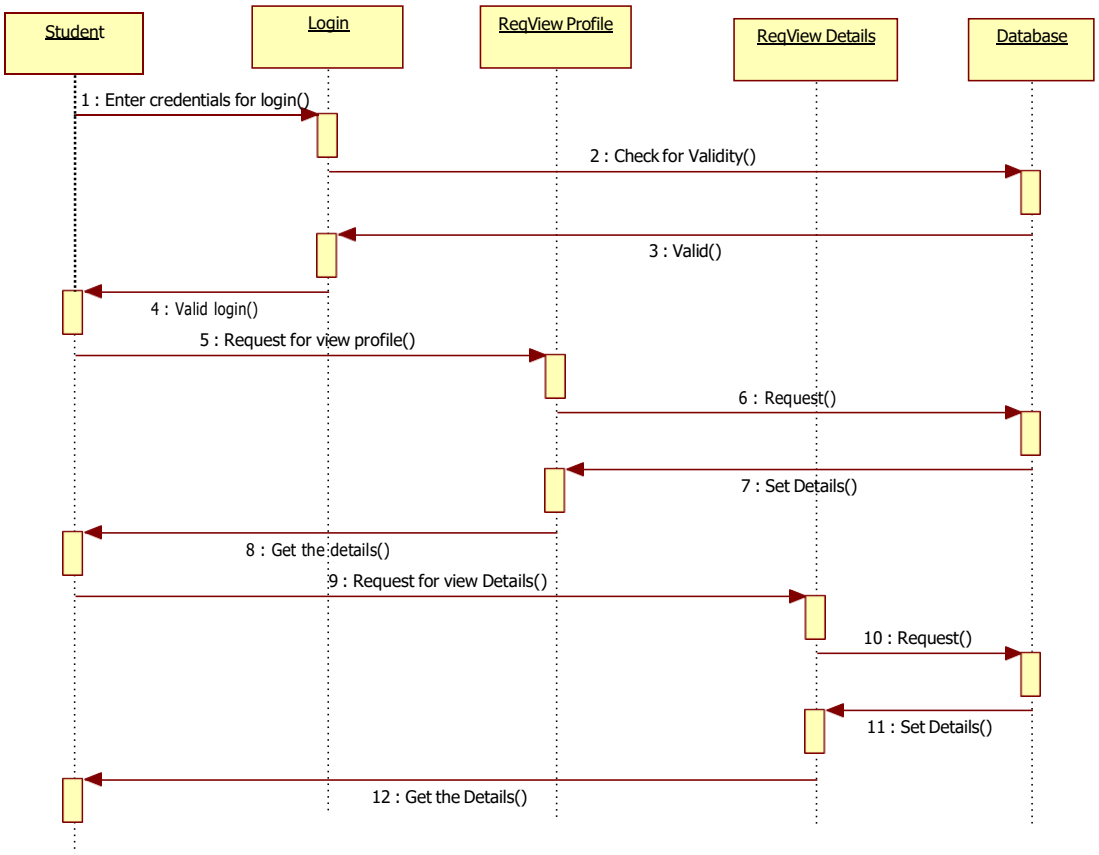
For Validity:



For Administrator:



For Student:



DOCUMENTATION OF SEQUENCE DIAGRAM

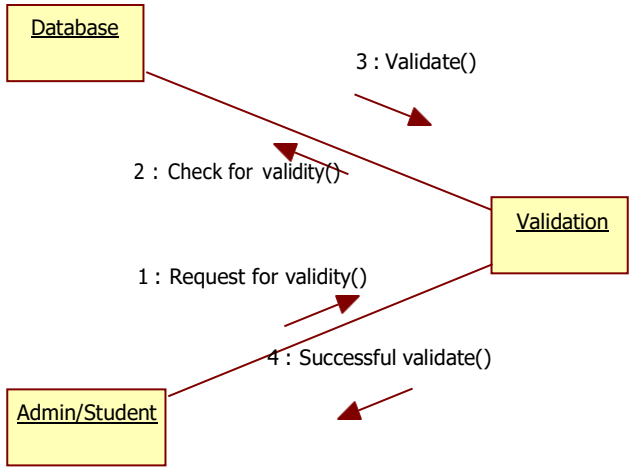
The sequence diagram describes the sequence of steps to show

a. The Admin login and registering for add Student Details.
b. The verification done by the interface and sending acknowledgement for registration.
c. Searching the database with login and displaying it for maintenance.

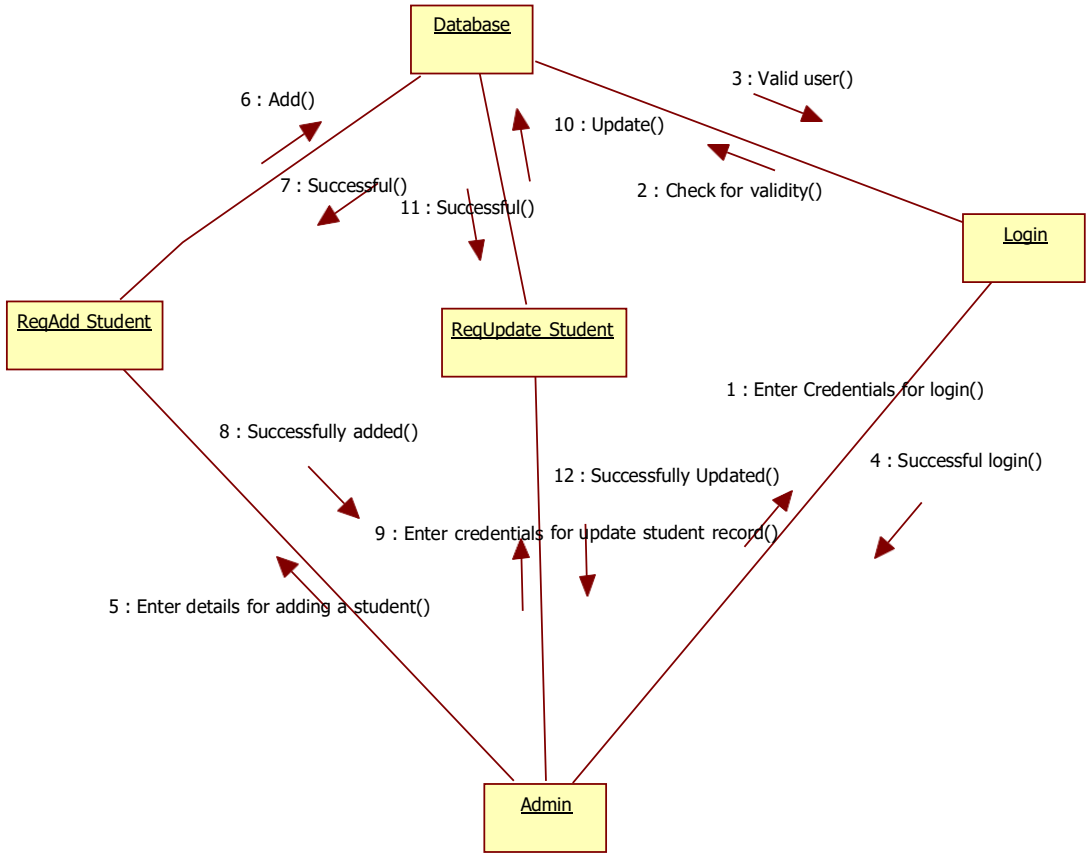
COLLABRATION DIAGRAM:

A collaboration diagram, also called a communication diagram or interaction diagram,. A sophisticated modeling tool can easily convert a collaboration diagram into a sequence diagram and the vice. A collaboration diagram resembles a flowchart that portrays the roles, functionality and behavior of individual objects as well as the overall operation of the system in real time

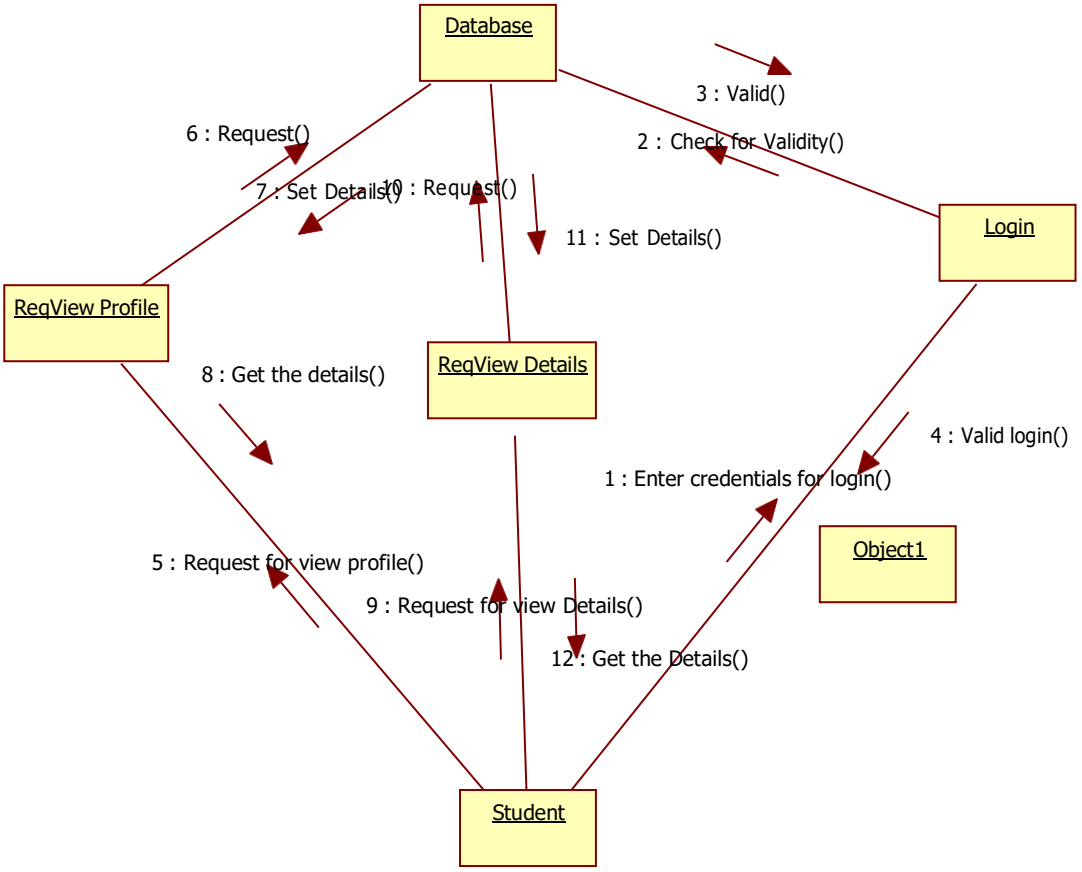
For Validity:



For Administrator:



For Student:



DOCUMENTATION OF COLLABRATION DIAGRAM

The collaboration diagram is to show how the Student registers and the authorities maintains the details of the registered students in the Information system. Here the sequence is numbered according to the flow of execution.

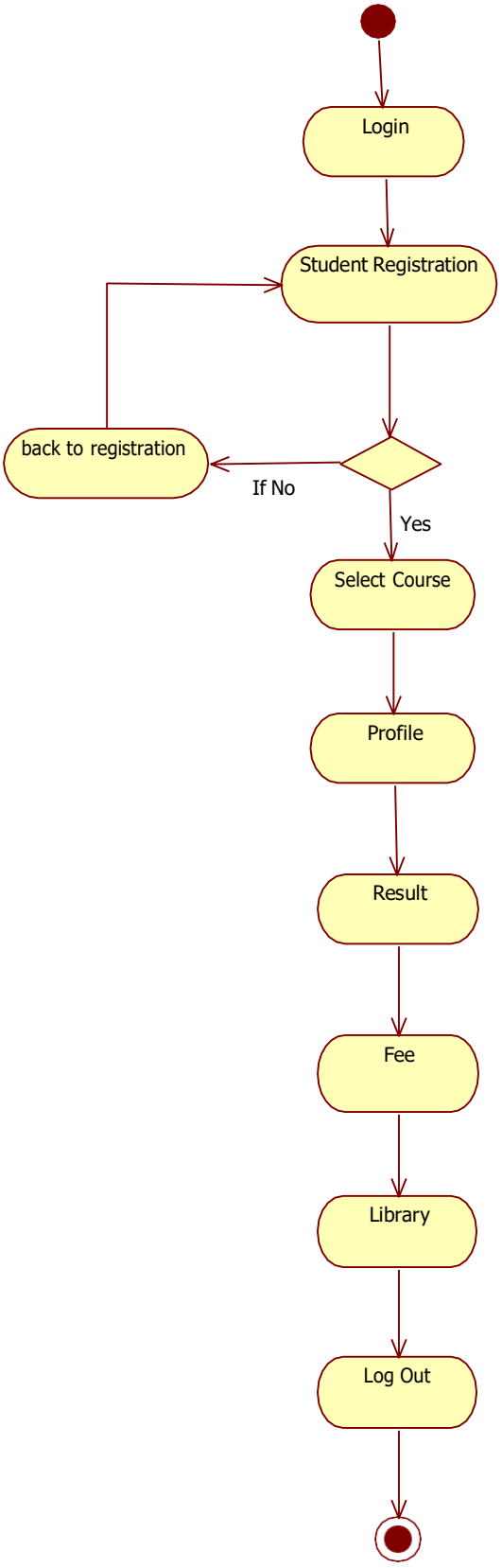
ACTIVITY DIAGRAM:

Activity diagrams are graphical representations of workflows of stepwise activities and actions with support for choice, iteration and concurrency. In the Unified Modeling Language, activity diagrams can be used to describe the business and operational step-by-step workflows of components in a system. An activity diagram shows the overall flow of control. An activity is shown as an rounded box containing the name of the operation.

DOCUMENTATION OF ACTIVITY DIAGRAM

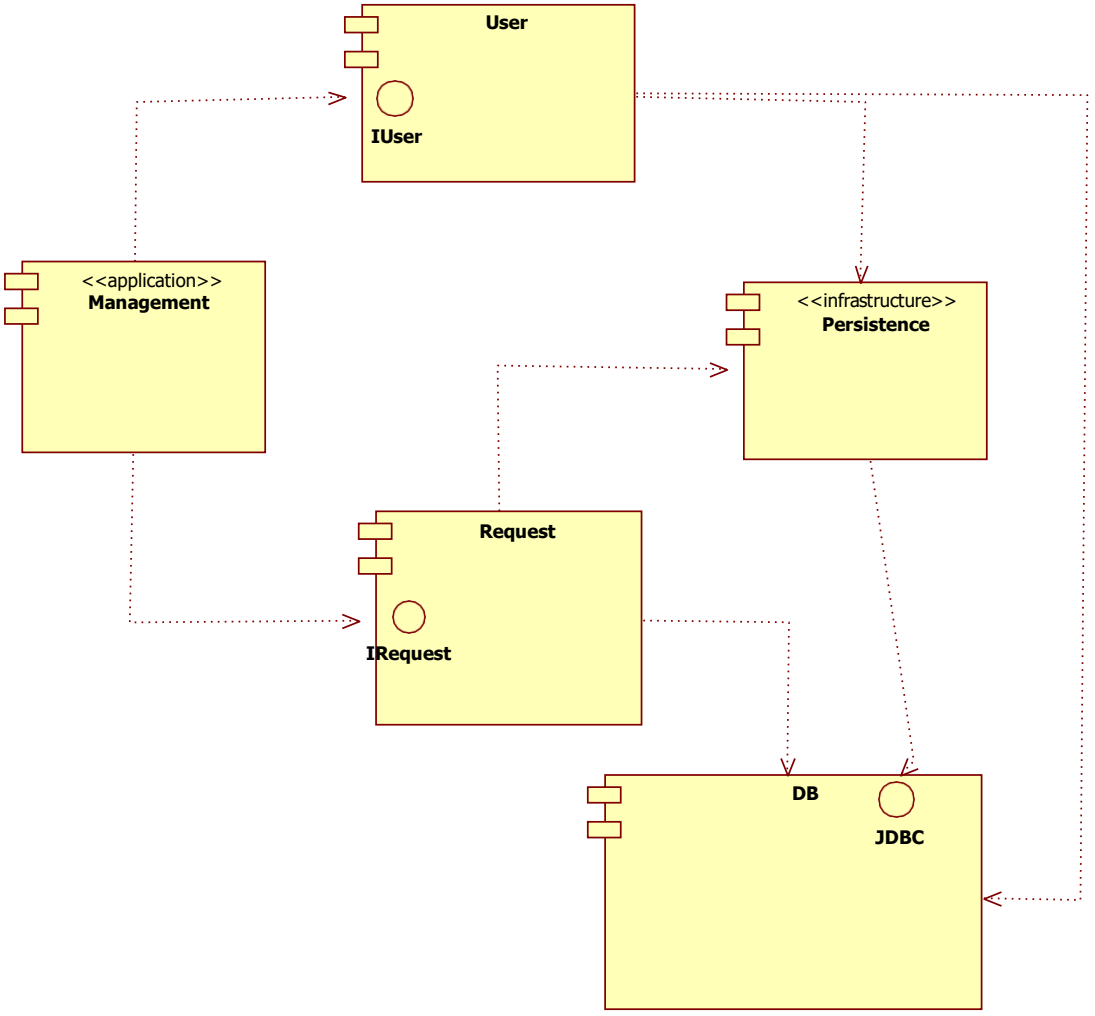
This activity diagram flow of stepwise activities performed in recruitment system.

a. The student details are Add and stored in database.
b. Select the course from the given Course by student.
c. Search Profile and Result with login and if data present in the database.
d. The searched data is displayed if available and then Log Out.


Login
Student Registration
back to registration
If No
Yes
Select Course
Profile
Result
Fee
Library
Log Out

COMPONENT DIAGRAM:

The component diagram's main purpose is to show the structural relationships between the components of a system. It is represented by boxed figure. Dependencies are represented by communication association.

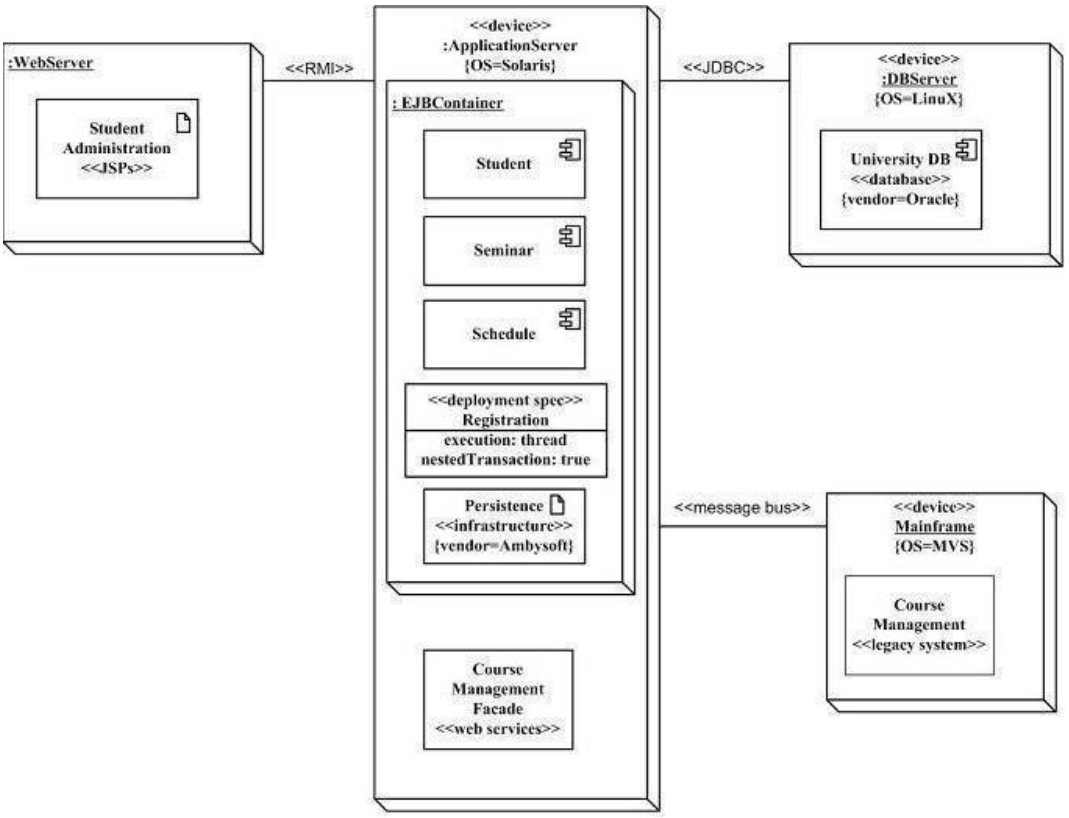


DOCUMENTATION OF COMPONENT DIAGRAM

The main component in this component diagram is Student Information system. And register, User and Manage, Request details are the components comes under the main component.

DEPLOYMENT DIAGRAM:

A deployment diagram in the unified modeling language serves to model the physical deployment of artifacts on deployment targets. Deployment Diagrams show "the allocation of artifacts to nodes according to the Deployments defined between them. It is represented by 3-dimentional box. Dependencies are represented by communication association.

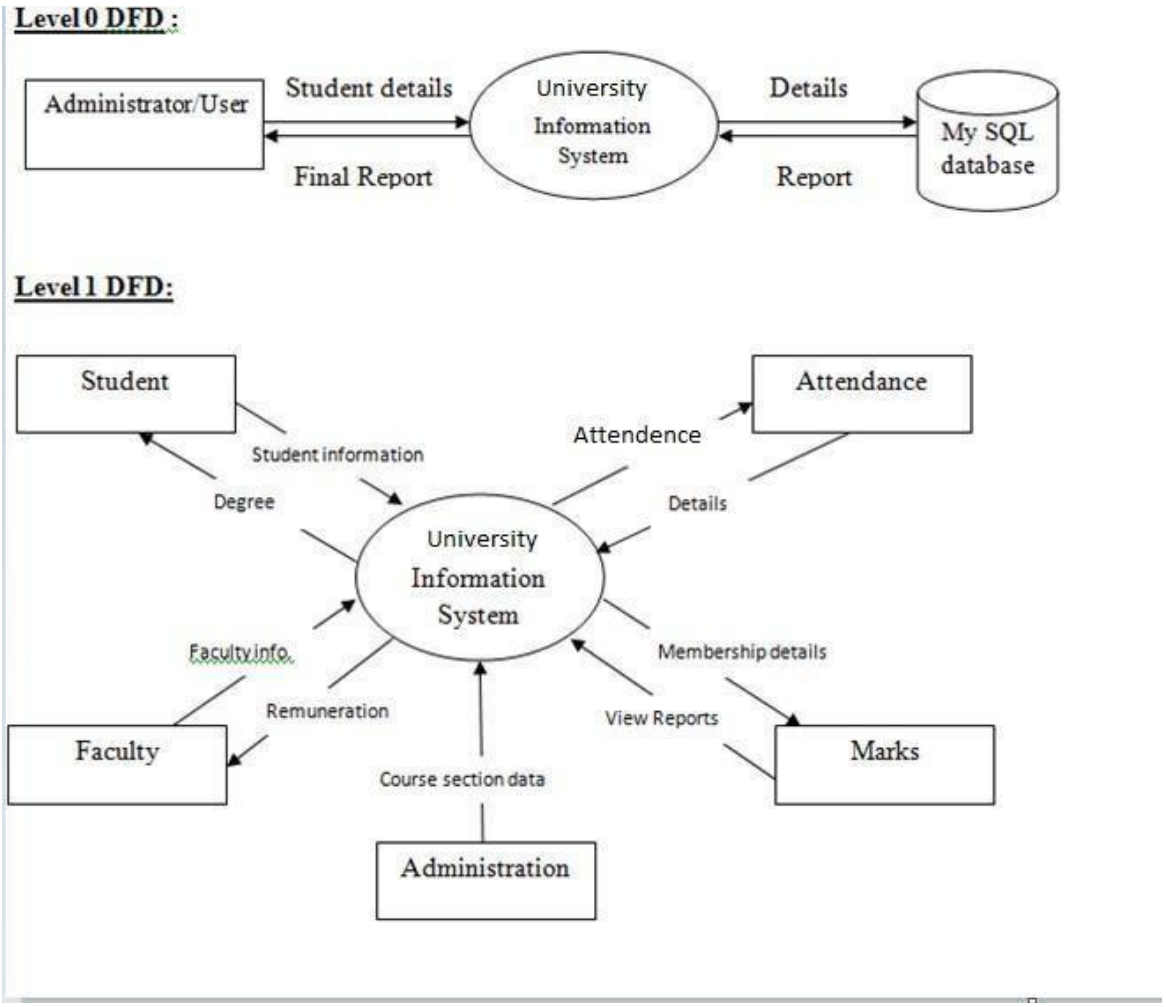


DOCUMENTATION OF DEPLOYMENT DIAGRAM

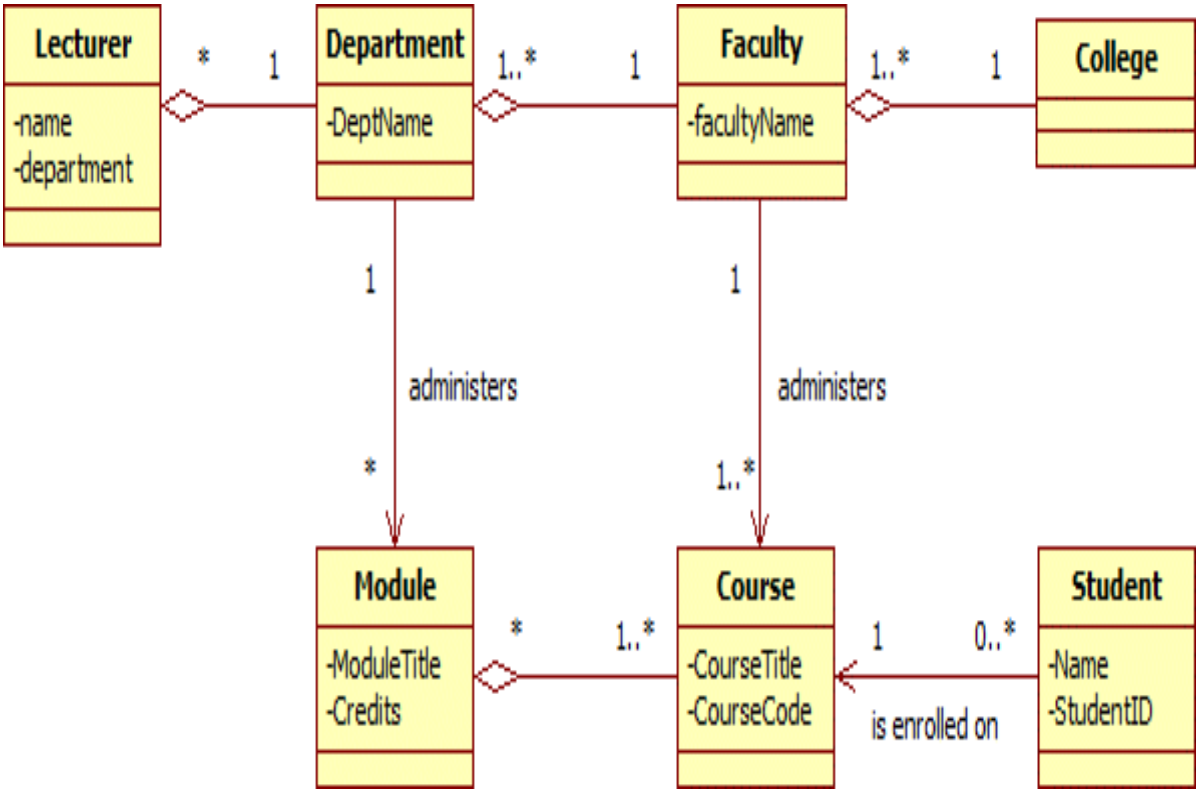
The processor in this deployment diagram is the College Information System which is the main part and the Student are the Admin, verify and search which are the some of the main activities performed in the system.

Data Flow diagram

A Data Flow Diagram (DFD) is a graphical representation of the "flow" of data through an information system, modeling its process aspects. A DFD is often used as a preliminary step to create an overview of the system, which can later be elaborated.



Object Diagram:



Result:

Ex. No.: 3	UML diagrams for Hostel Management system
Date:	

Aim:

To develop a project Hostel Management system using Enterprise Architecture Software and to implement the software in Java.

Description for Hostel Management System:

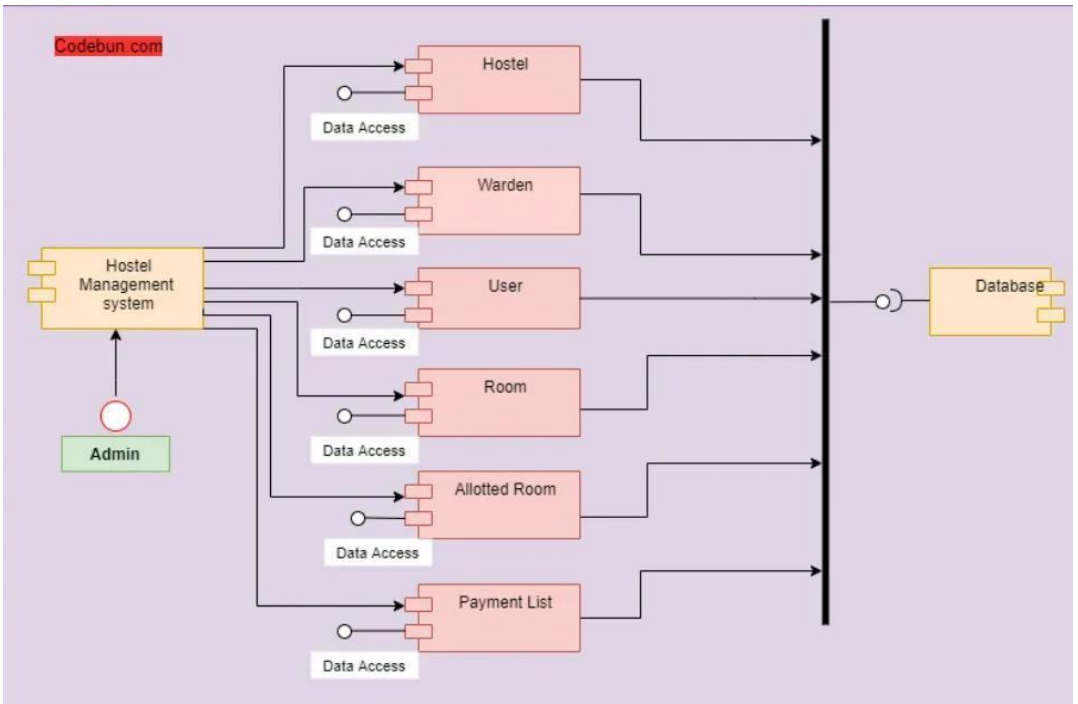
Hostel management gives an idea about student's details, room allocation, mess expenditure are maintained in the particular concern. The hostel management system also includes some special features. The administration has the unique identity for each members as well as students details. The stock management has also held by mess expenditure, the mess expenditure that used to calculate the mess bills of each of the students. The modules of this project are student details, attendance details, room details, mess modules.

Hostel Management System provides the following services:

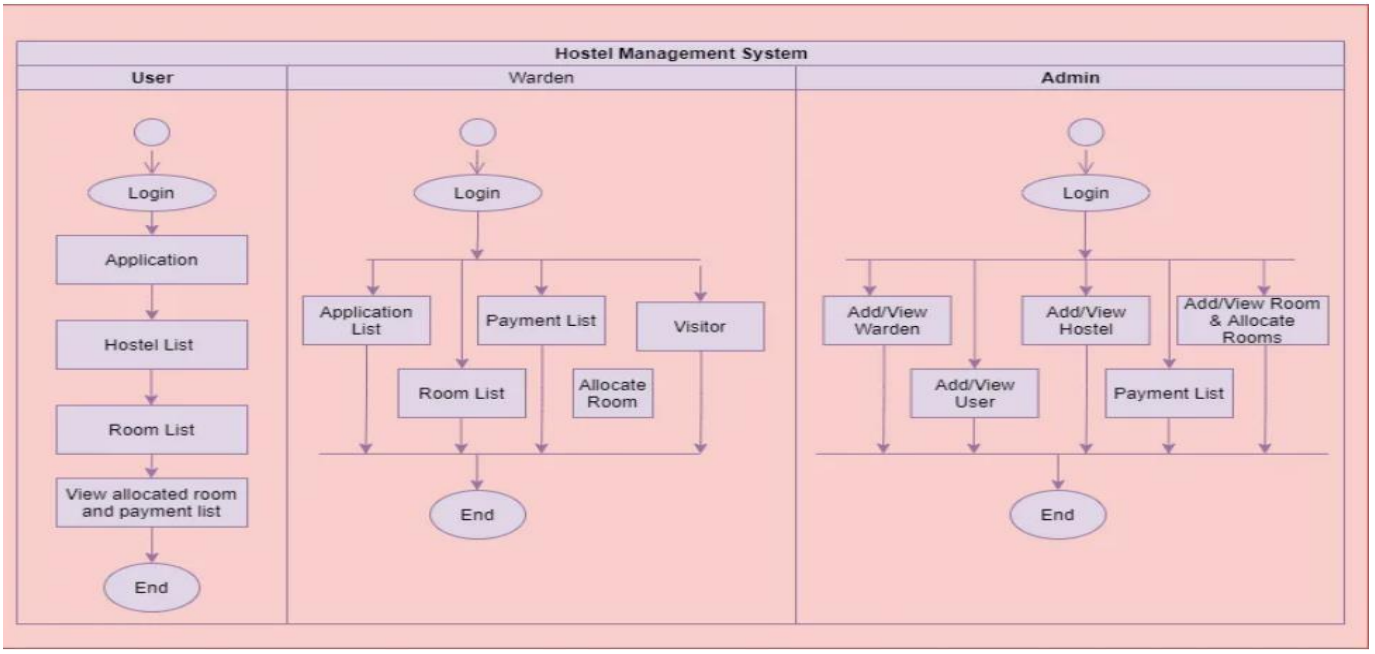
- Hostel Management System able to provide student details.
- Hostel Management System able to provide room details of student.
- Hostel Management System able to provide mess expenditure of student.

Use case Diagram:

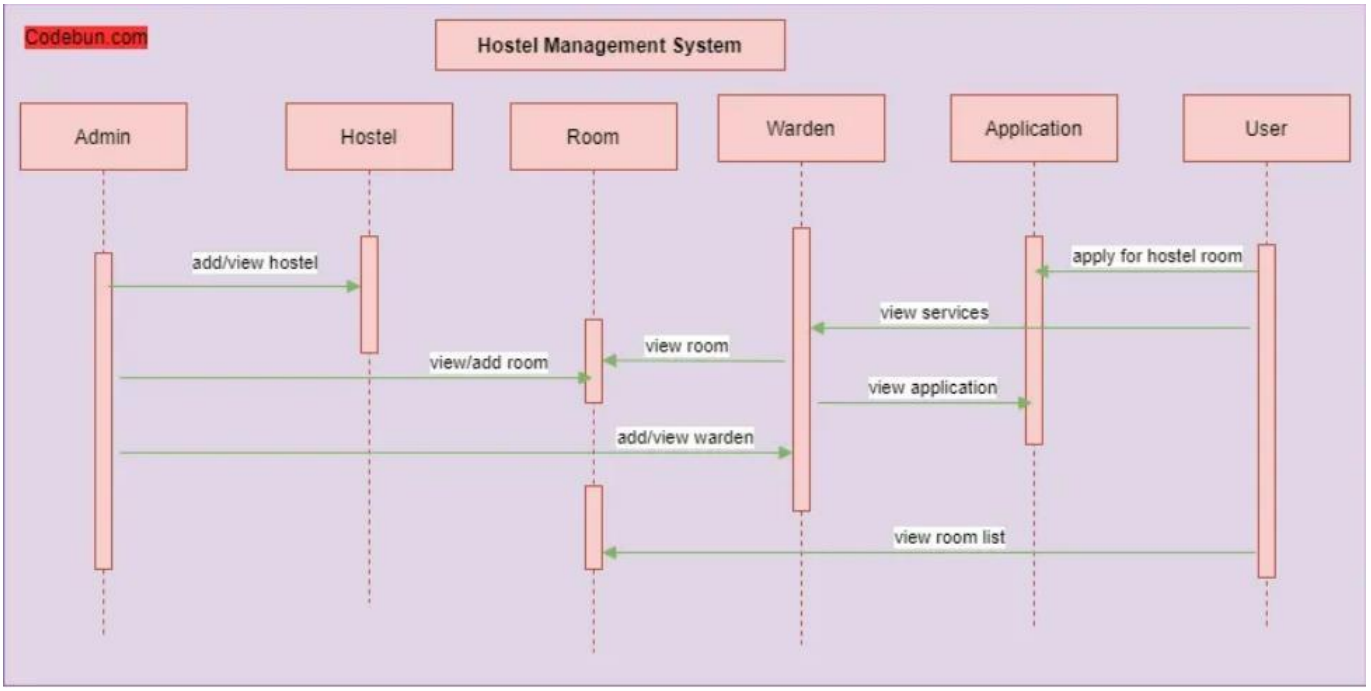
Component diagram for Hostel Management Project



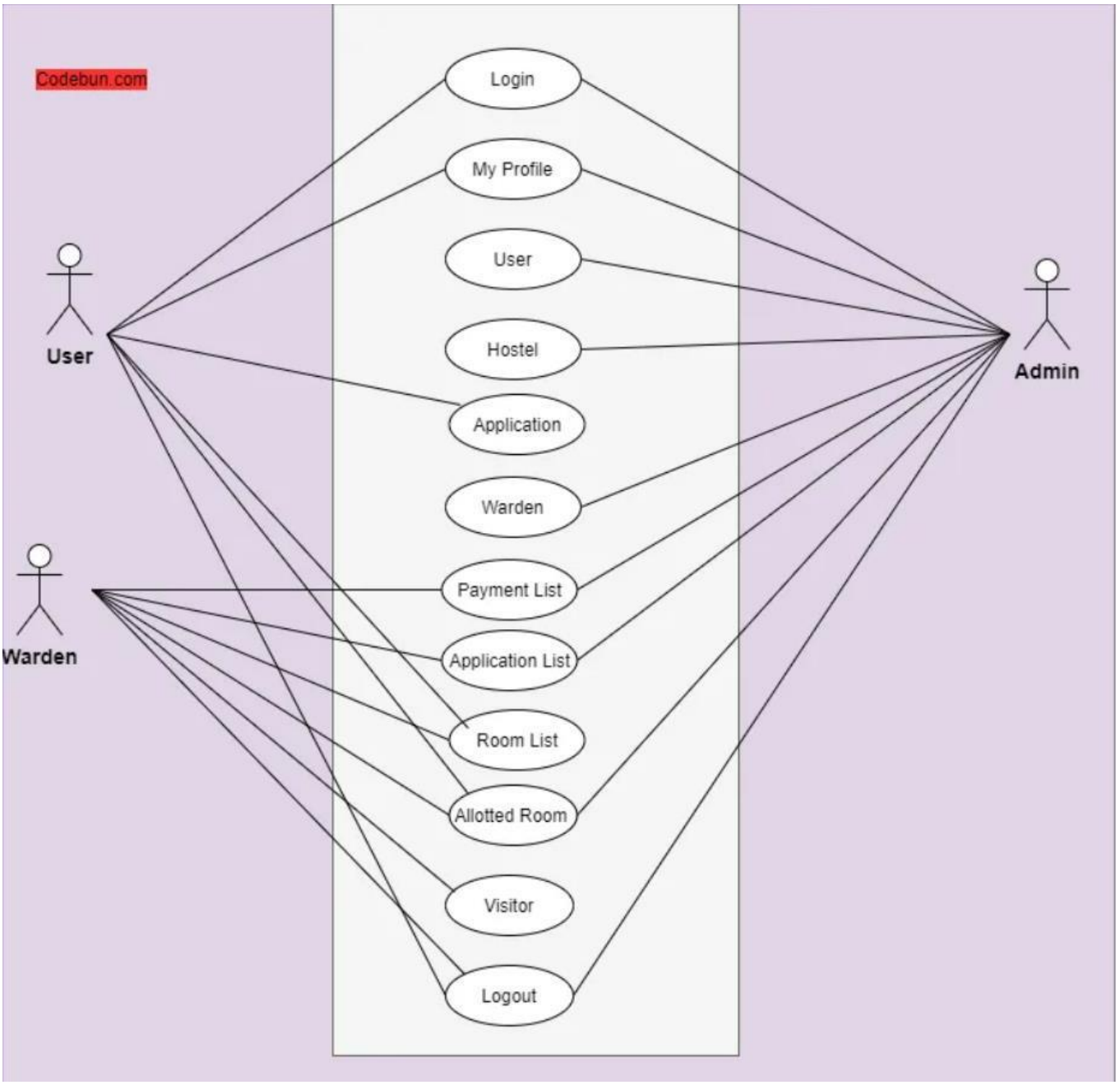
Activity diagram for Hostel Management Project



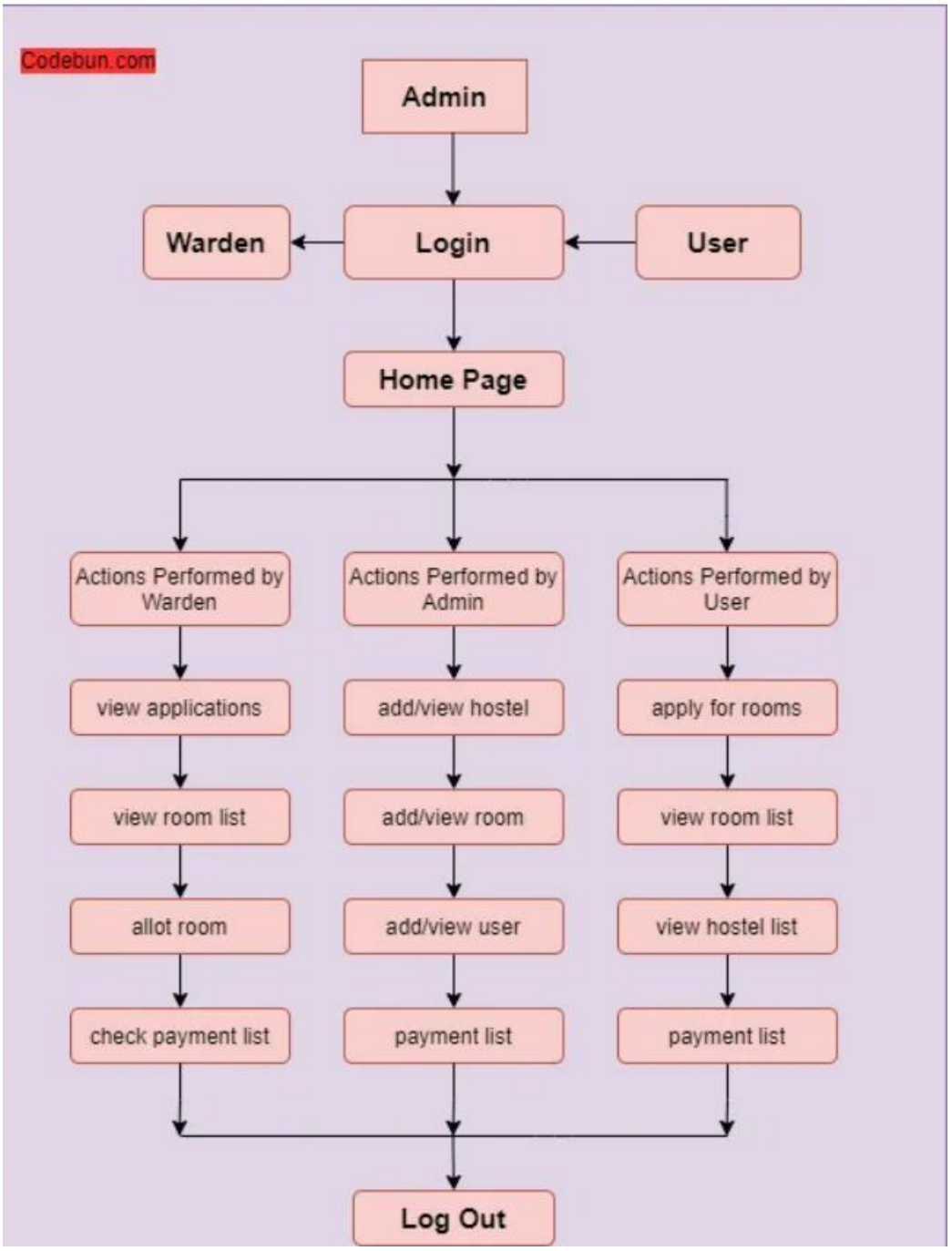
Sequence diagram for Hostel Management Project



Use Case Diagram for Hostel Management Project



UML DFD Diagram for Hostel Management Project



Ex. No.: 4	UML diagrams for ATM system
Date:	

Aim:

To develop a project ATM system using Enterprise Architecture Software and to implement the software in Java.

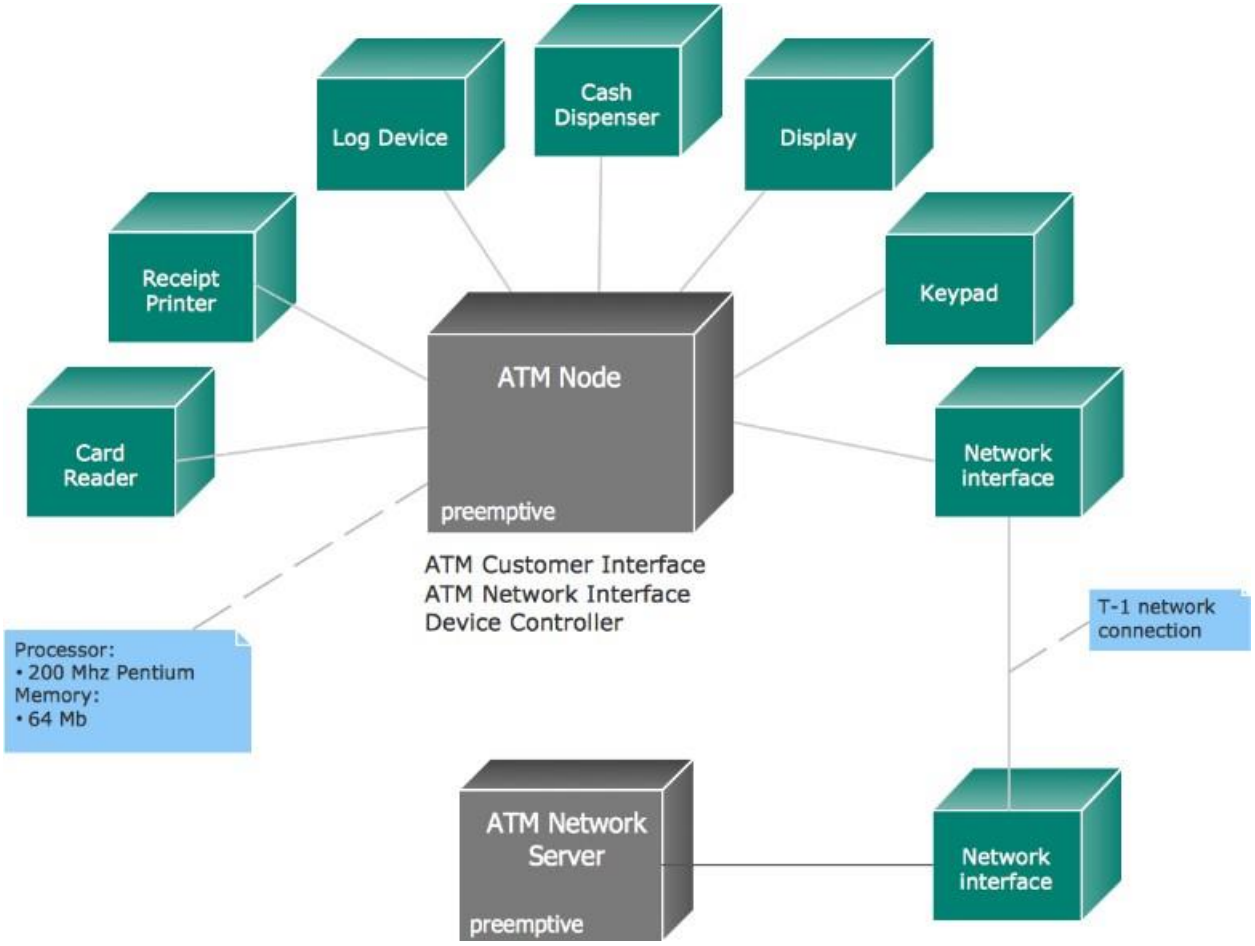
Description for ATM:

The software to be designed will control a simulated Automated Teller Machine (ATM) having a magnetic stripe reader for reading an ATM card, a customer console (keyboard and display) for interaction with the customer, a slot for depositing envelopes, a dispenser for cash (in multiples of Rs. 100, Rs. 500 and Rs. 2000), a printer for printing customer receipts, and a key-operated switch to allow an operator to start or stop the machine. The ATM will communicate with the bank's computer over an appropriate communication link. The ATM will service one customer at a time. A customer will be required to insert an ATM card and enter a personal identification number (PIN) - both of which will be sent to the bank for validation as part of each transaction. The customer will then be able to perform one or more transactions.

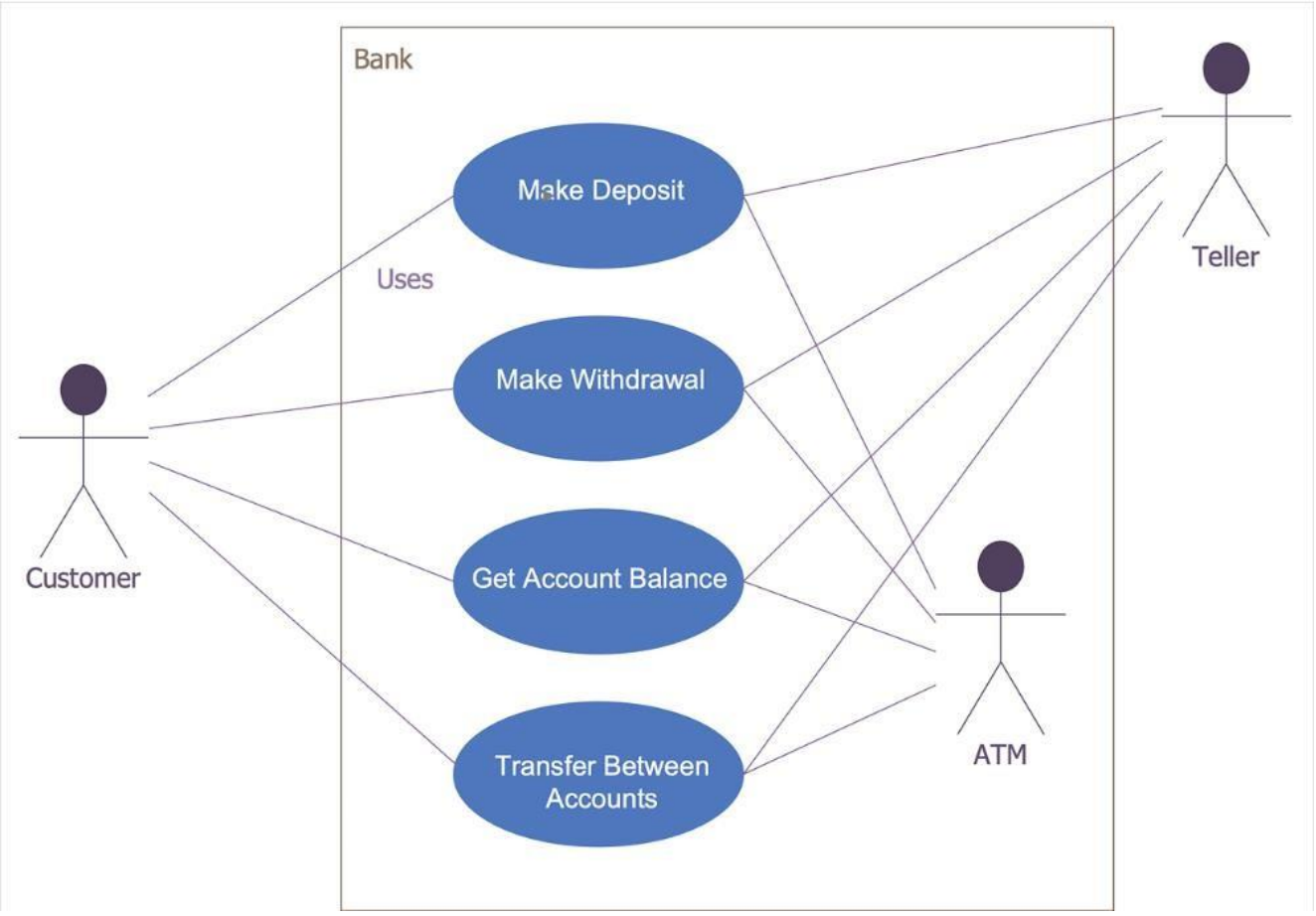
The ATM will communicate each transaction to the bank and obtain verification that it was allowed by the bank. Ordinarily, a transaction will be considered complete by the bank once it has been approved. If the bank determines that the customer's PIN is invalid, the customer will be required to re-enter the PIN before a transaction can proceed. If the customer is unable to successfully enter the PIN after three tries, the card will be permanently retained by the machine, and the customer will have to contact the bank to get it back If a transaction fails for any reason other than an invalid PIN, the ATM will display an explanation of the problem, and will then ask the customer whether he/she wants to do another transaction.

Use case Diagram:

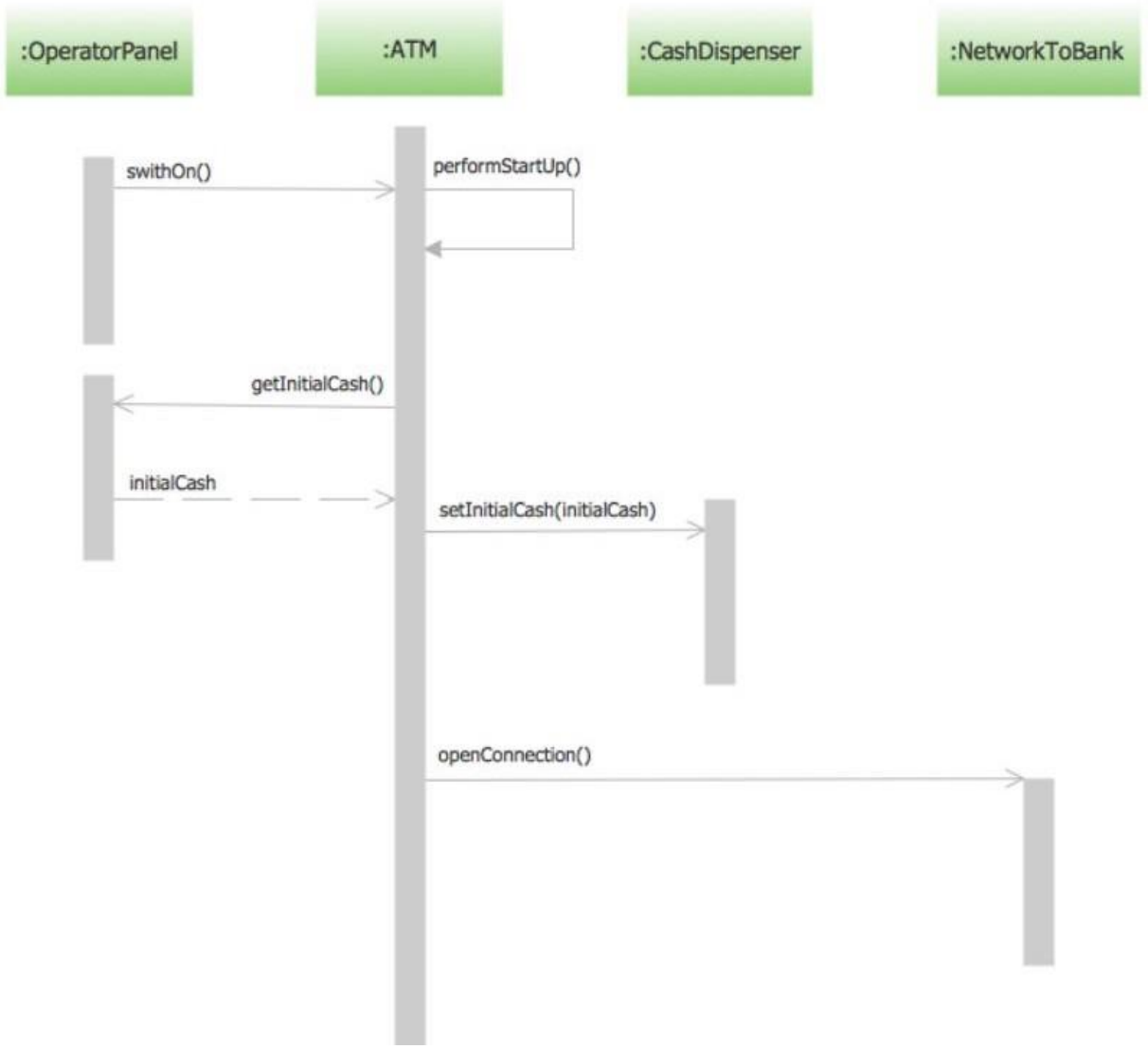
UML Deployment Diagram Example - ATM System UML diagrams

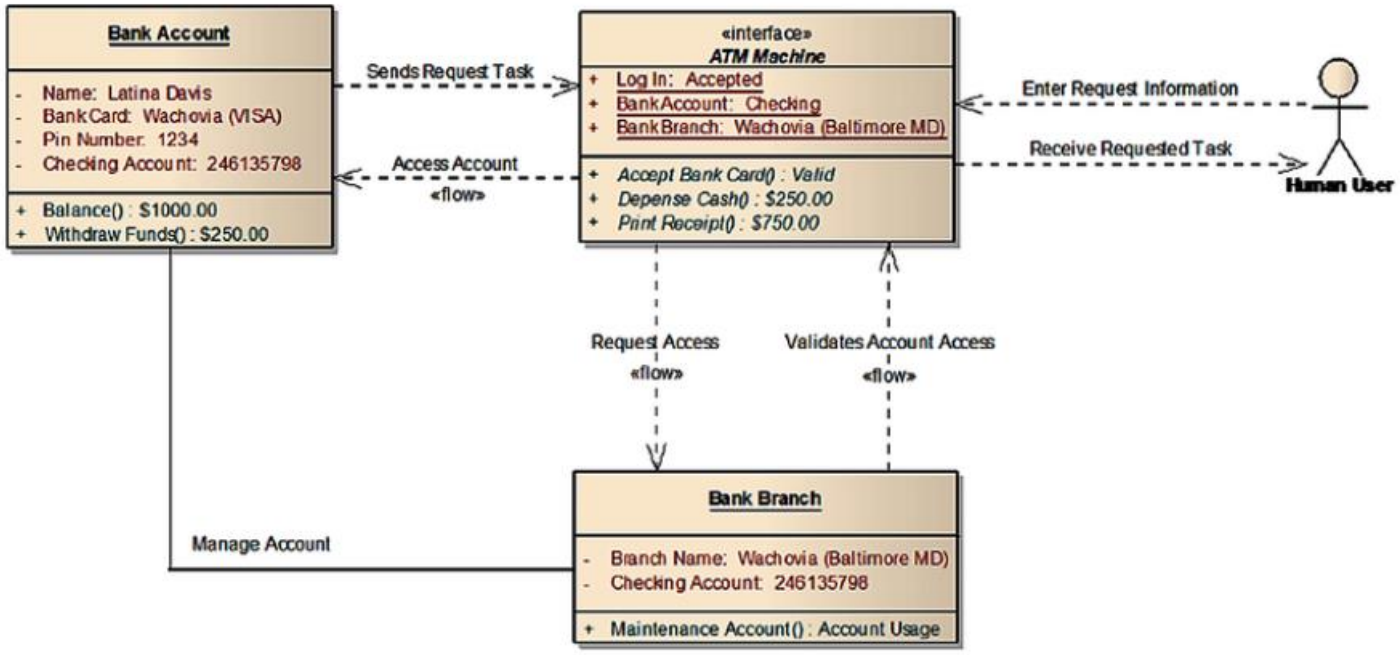


ATM Use Case Diagram



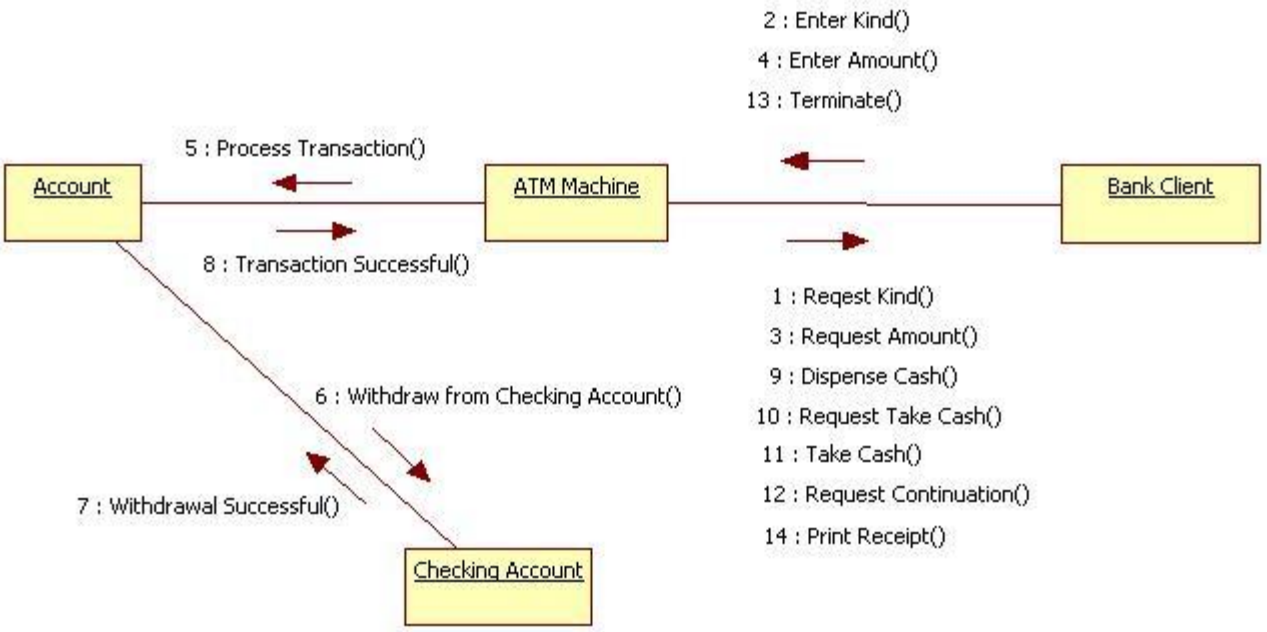
Sequence UML for ATM





ATM systems object diagram

___ATM collaboration diagram



Ex. No.: 5	**UML diagrams for Library management system**
Date:	

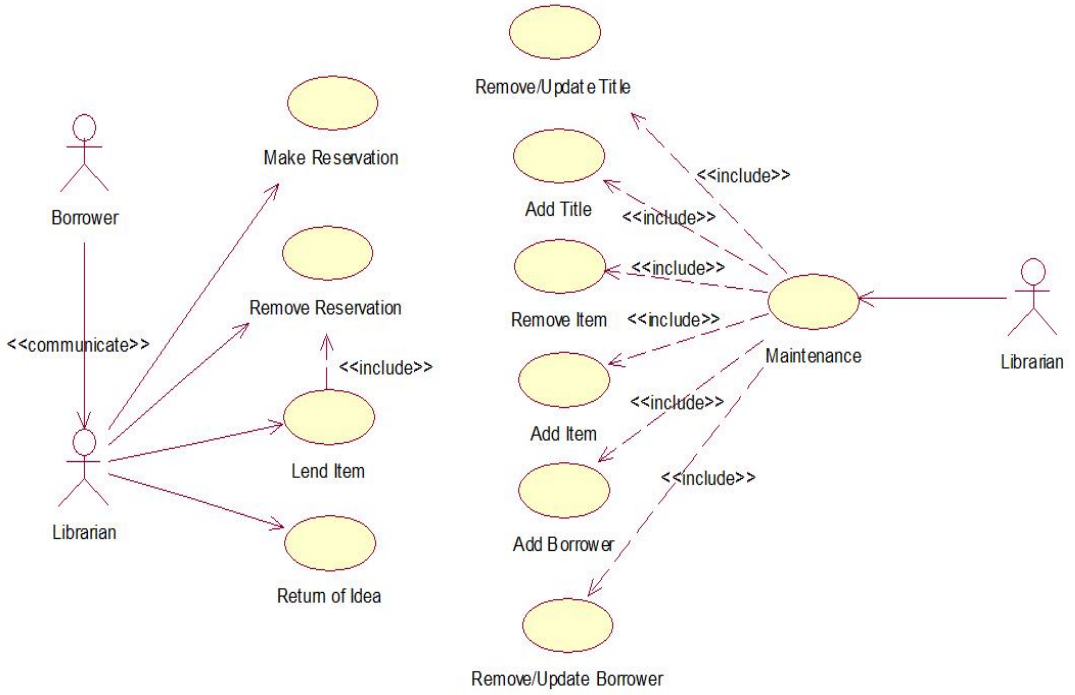
Aim:

To develop a project Library management system using Enterprise Architecture Software and to implement the software in Java.

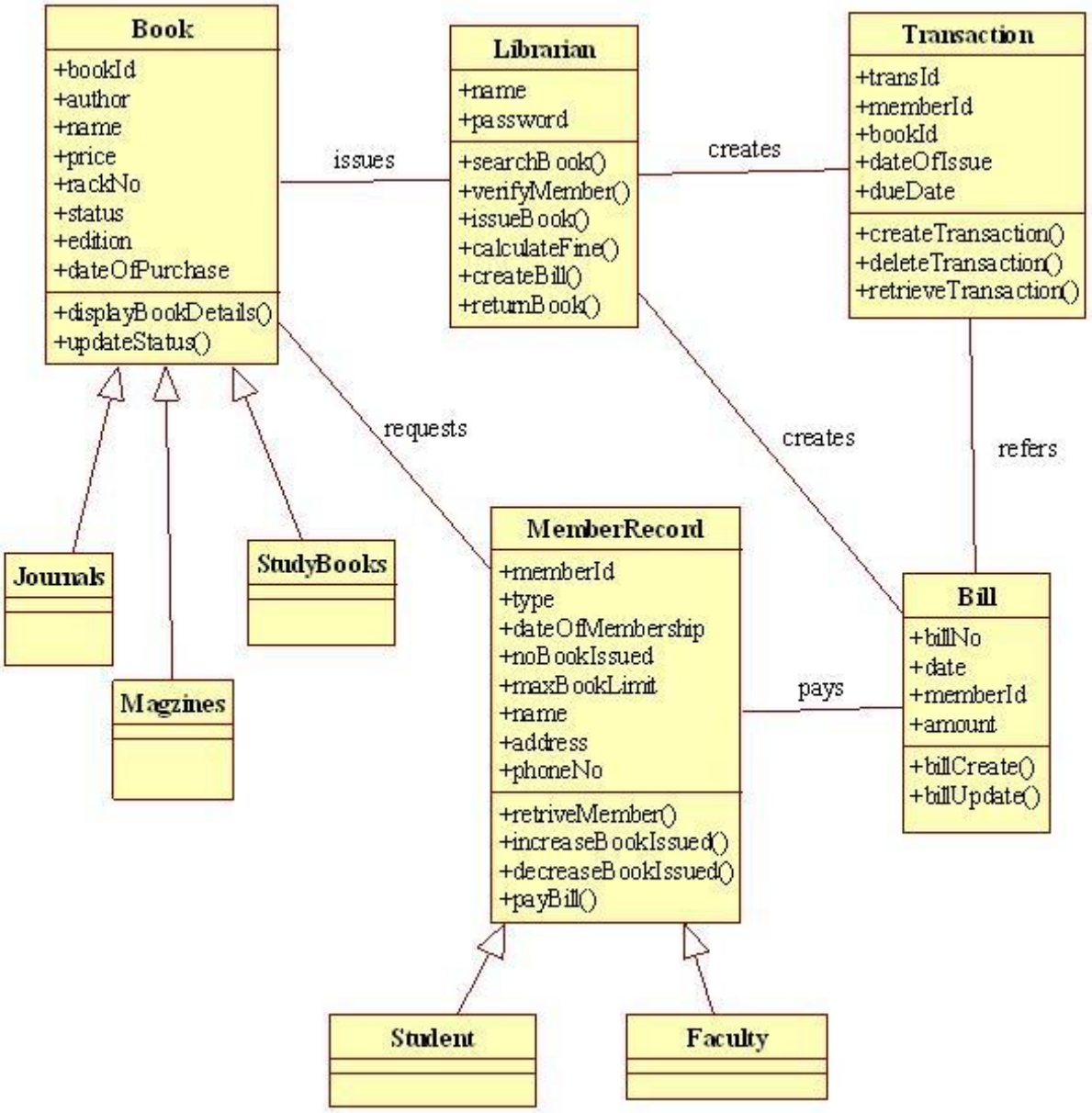
Description:

The library management system is a software system that issues books and magazines to registered students only. The student has to login after getting registered to the system. The borrower of the book can perform various functions such as searching for desired book, get the issued book and return the book.

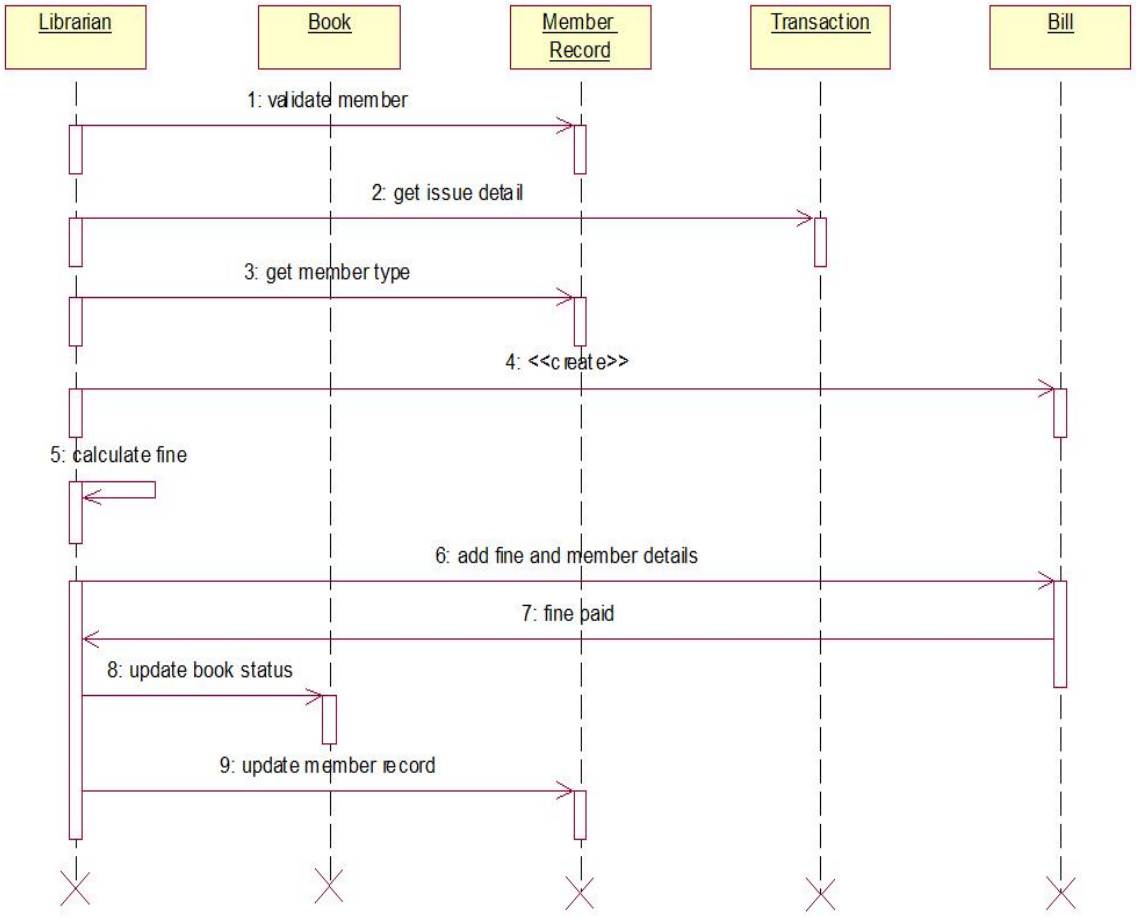
Use case Diagram:



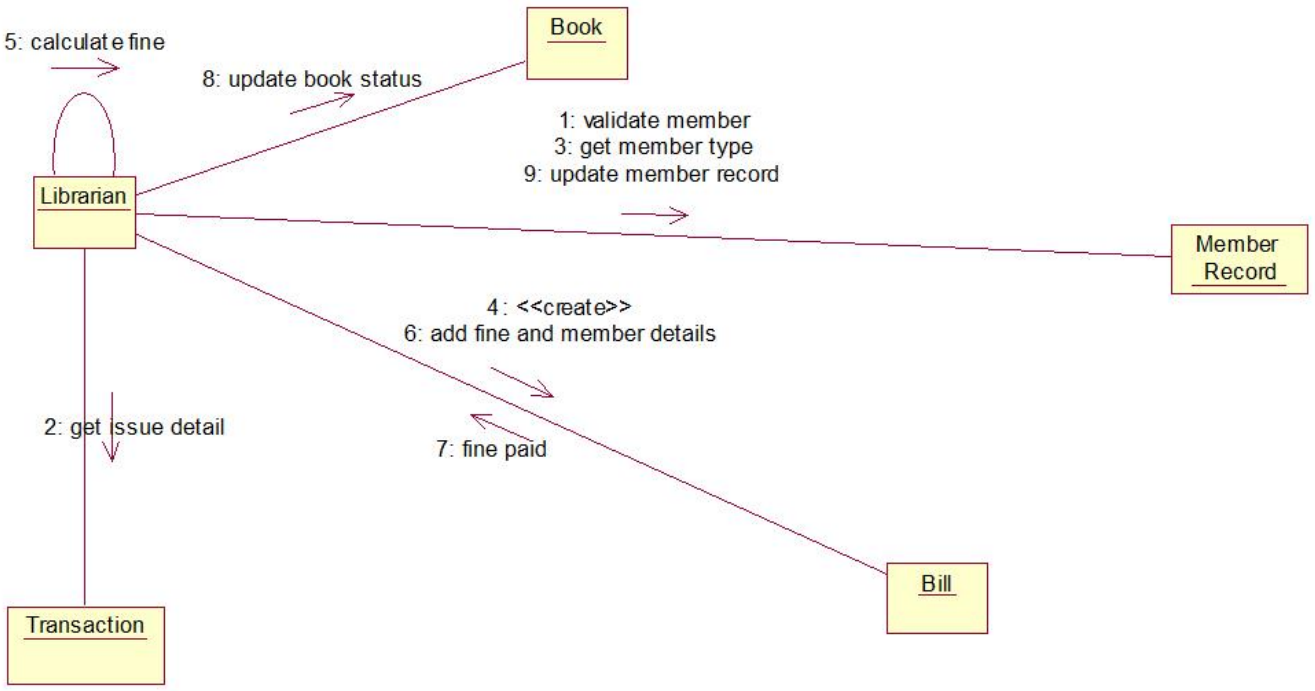
Class Diagram:



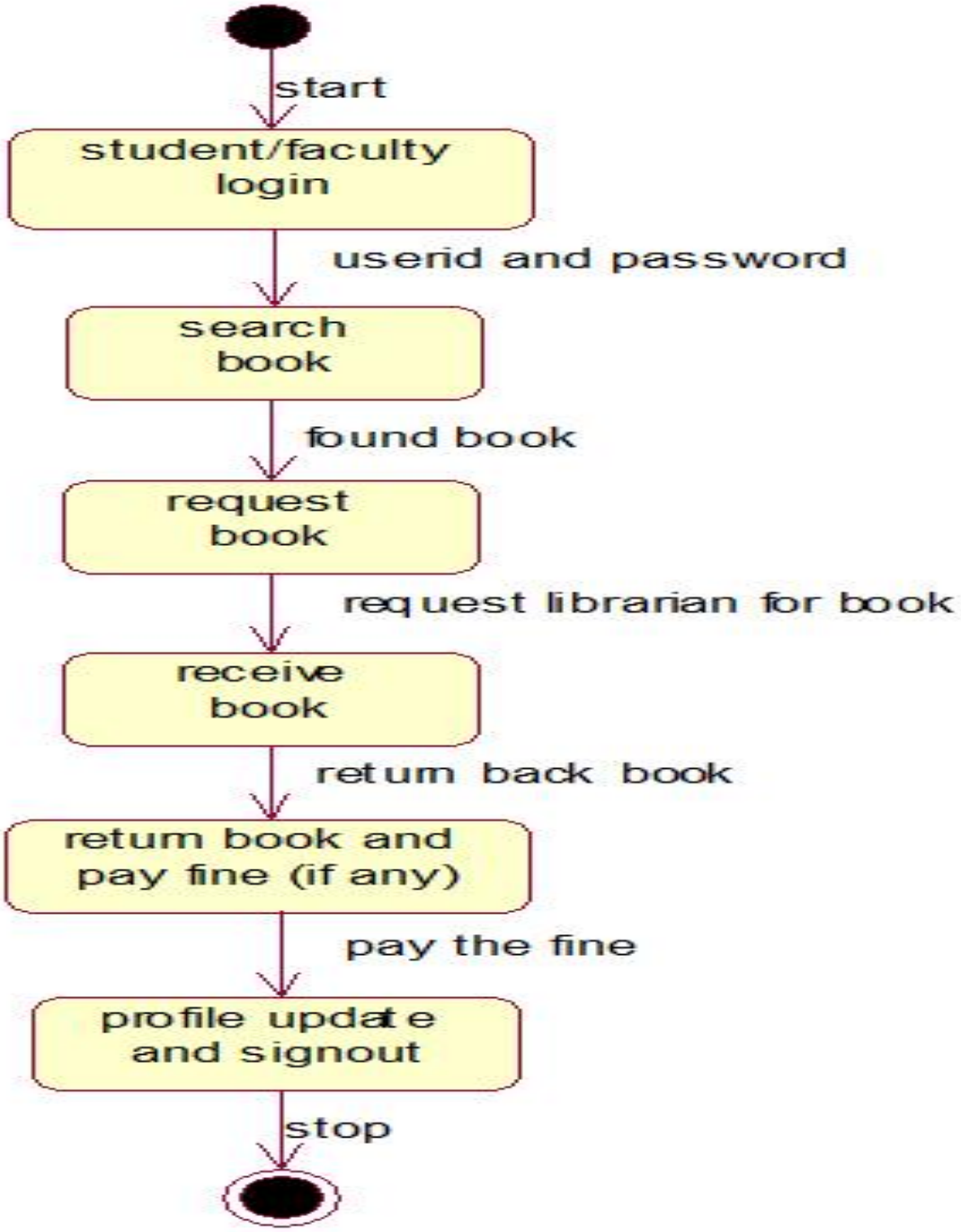
Sequence Diagram:



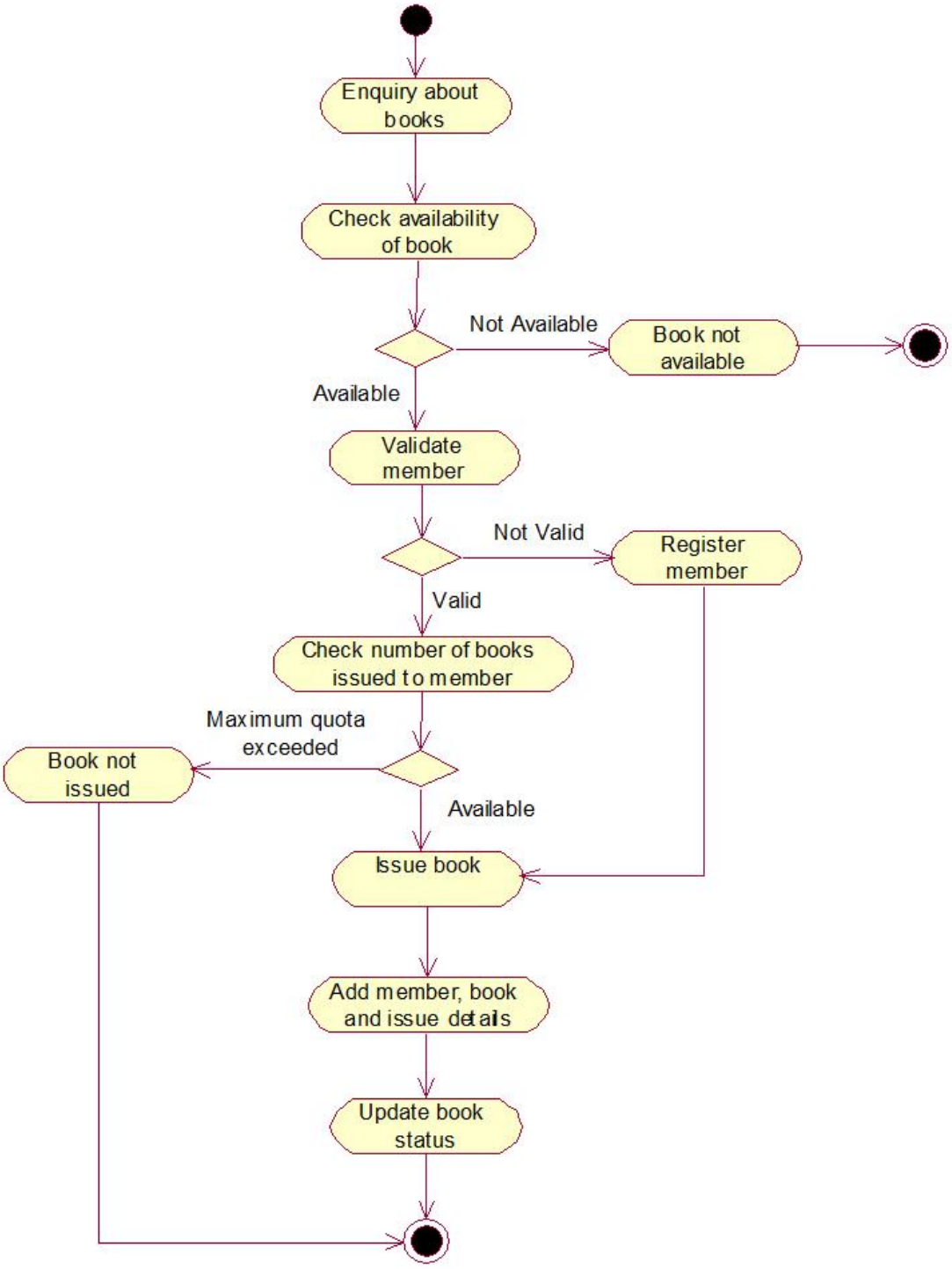
Collaboration Diagram:



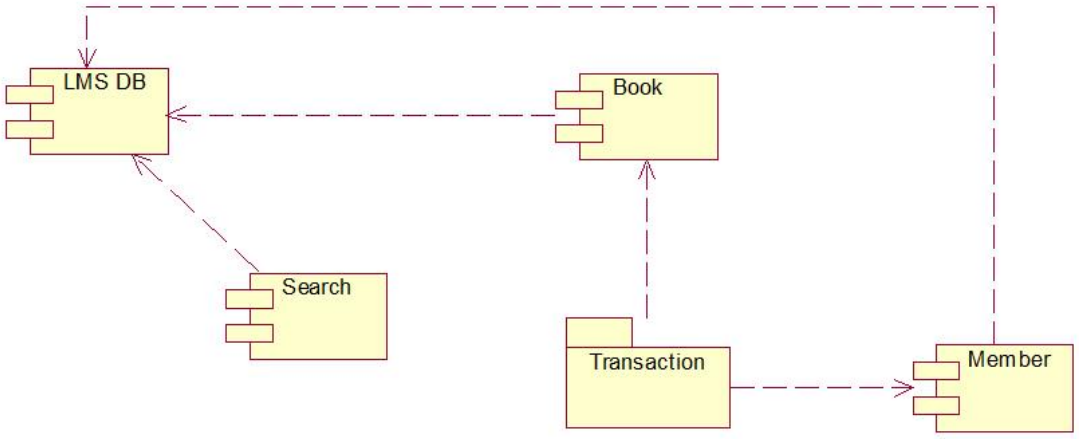
State Chart Diagram:



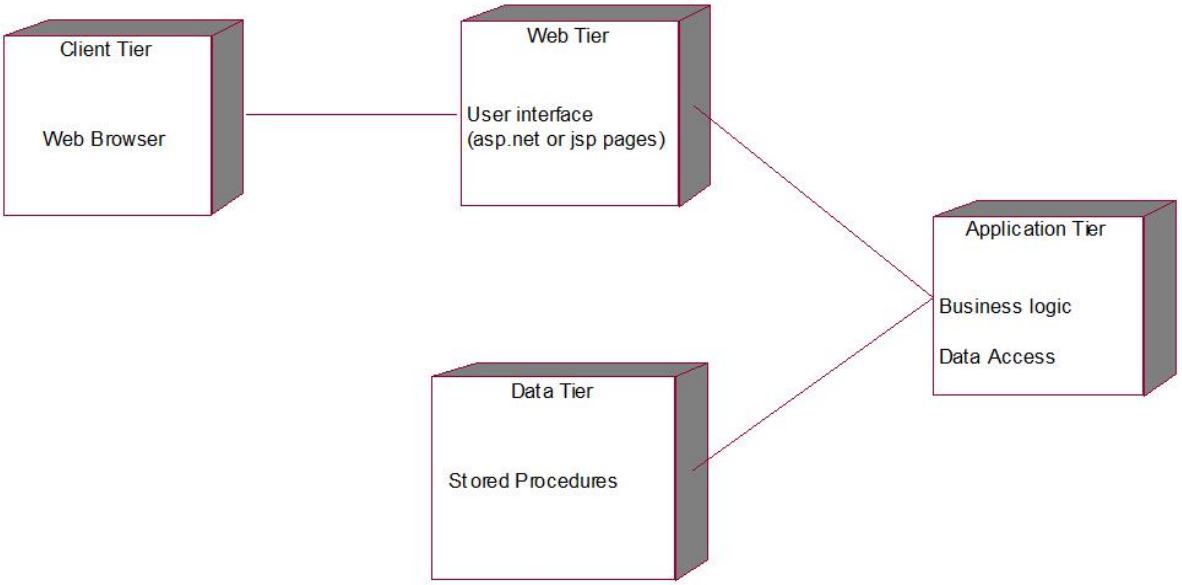
Activity Diagram:



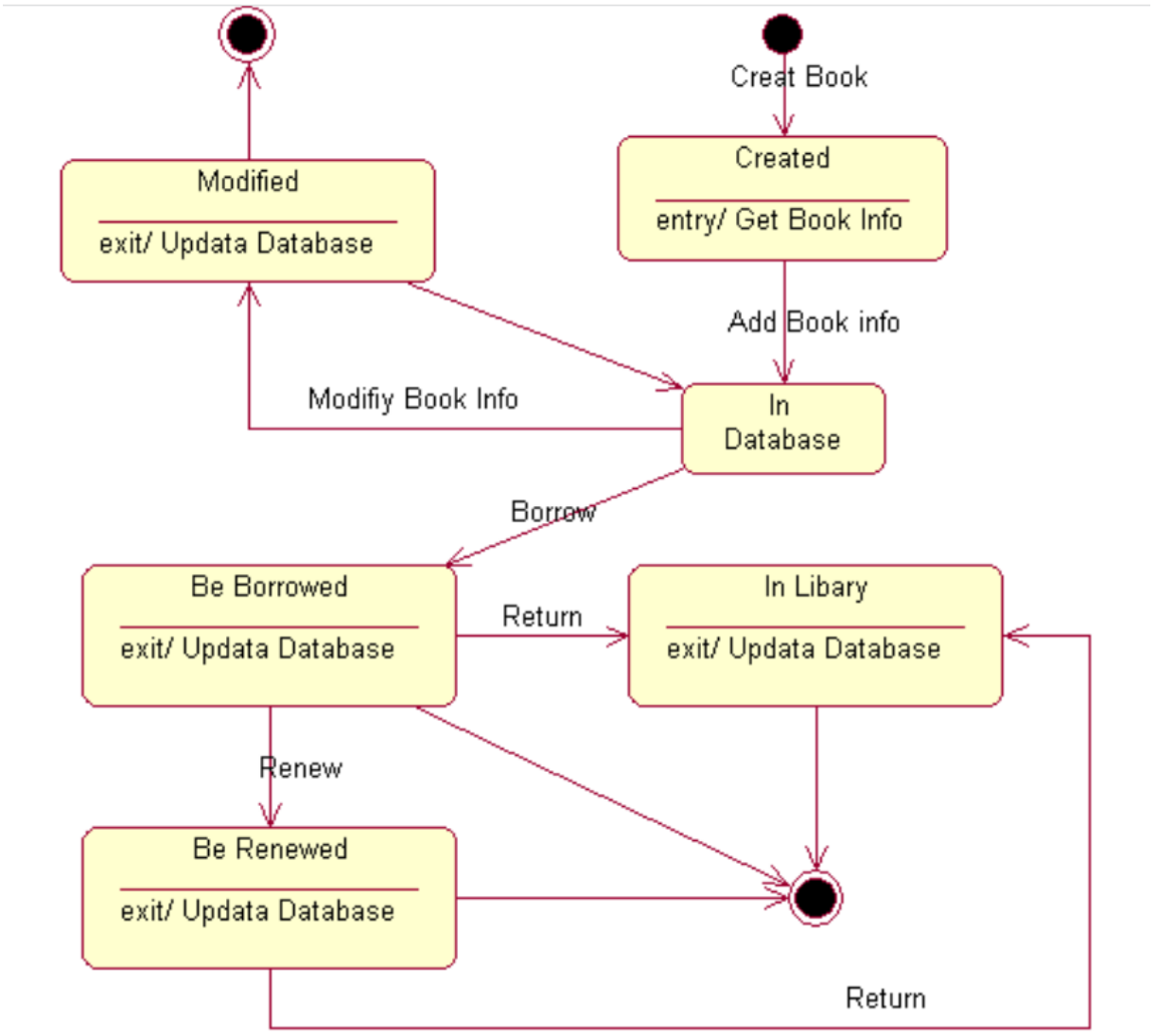
Component Diagram:



Deployment Diagram:



Object Diagram:



Result:

Ex. No.: 6	**UML diagrams for Passport Automation System**
Date:	

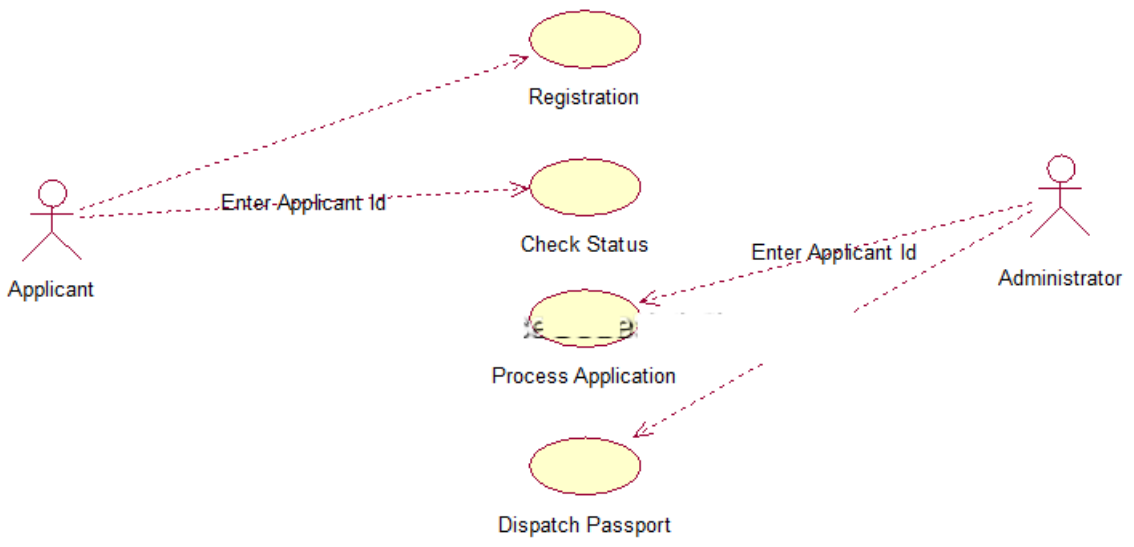
Aim:

To Analyze the Problems, Gather the SRS(Software Requirements Specification) and Model the UML or OOAD diagrams for "Passport Automation System

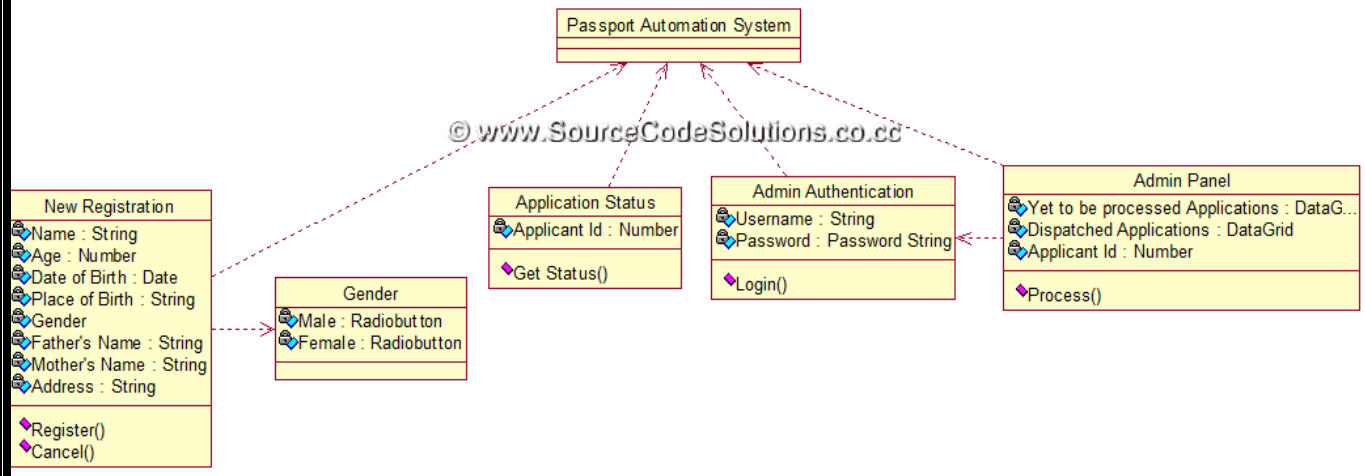
Description:

Passport Automation System is used in the effective dispatch of passport to all of the applicants. This system adopts a comprehensive approach to minimize the manual work and schedule resources, time in a cogent manner. The core of the system is to get the online registration form (with details such as name, address etc.,) filled by the applicant whose testament is verified for its genuineness by the Passport Automation System with respect to the already existing information in the database.

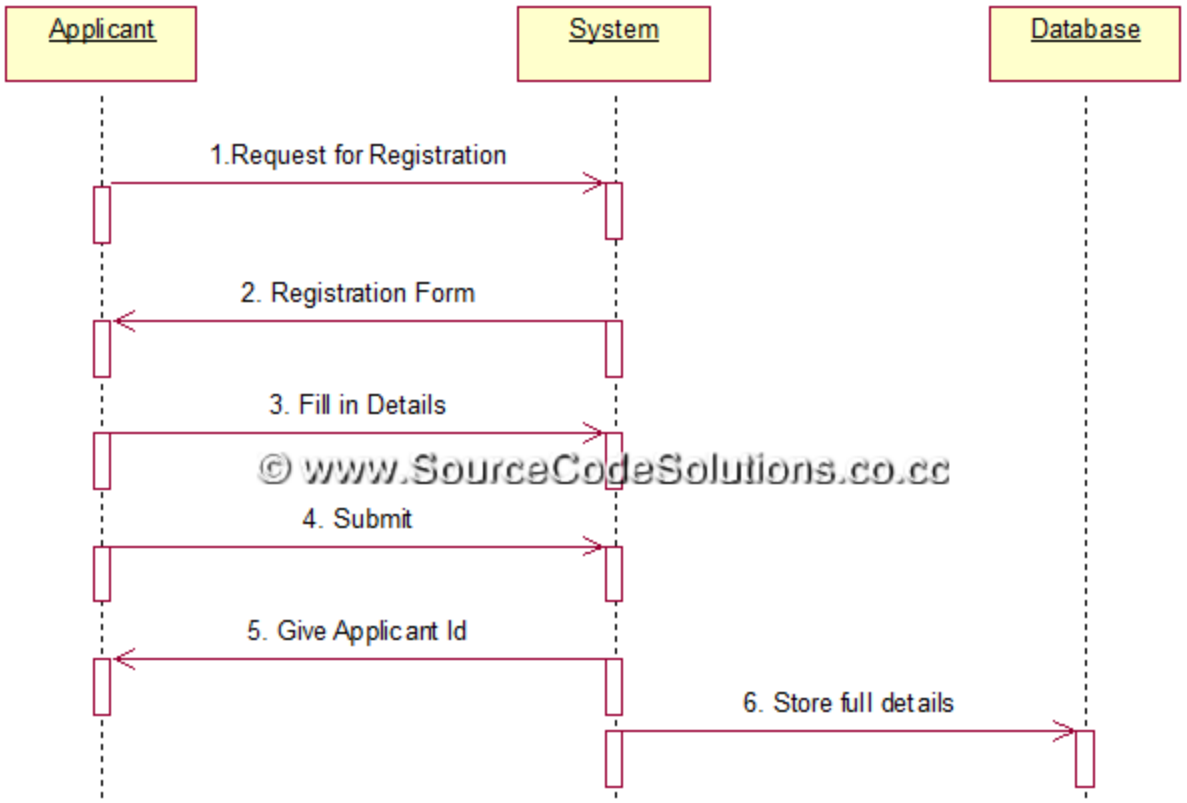
Use case Diagram:



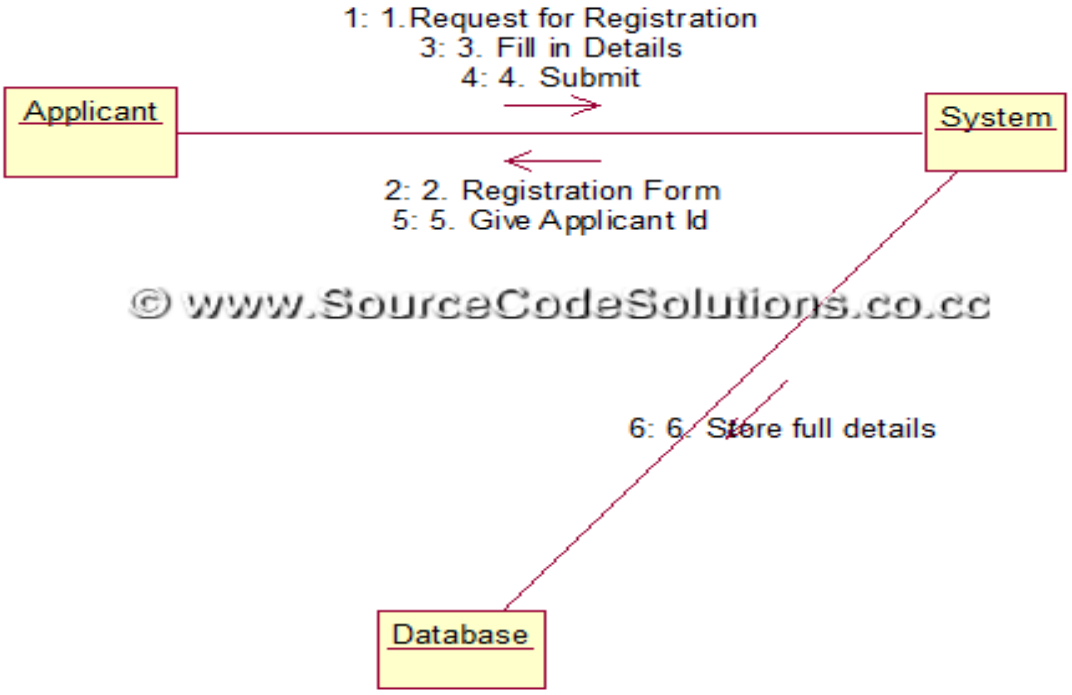
Class Diagram:



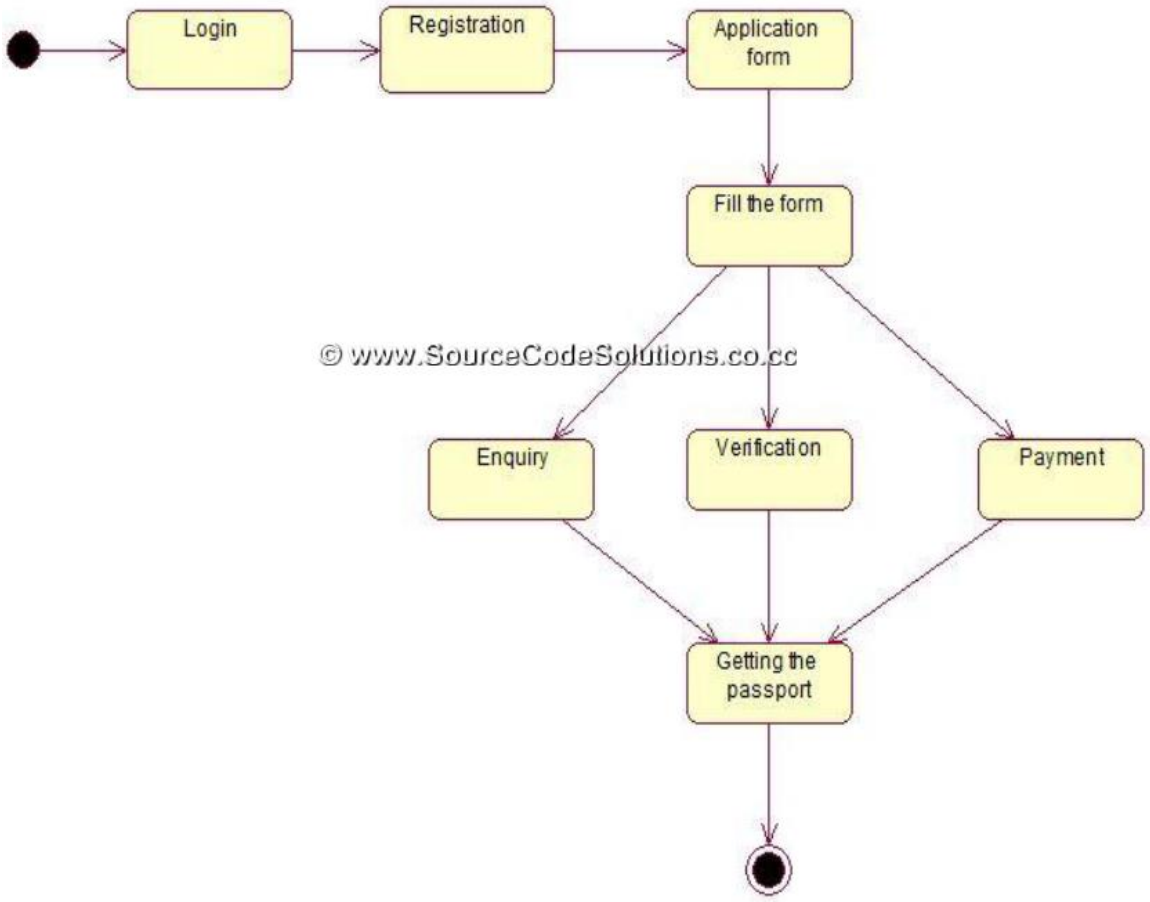
Sequence Diagram:



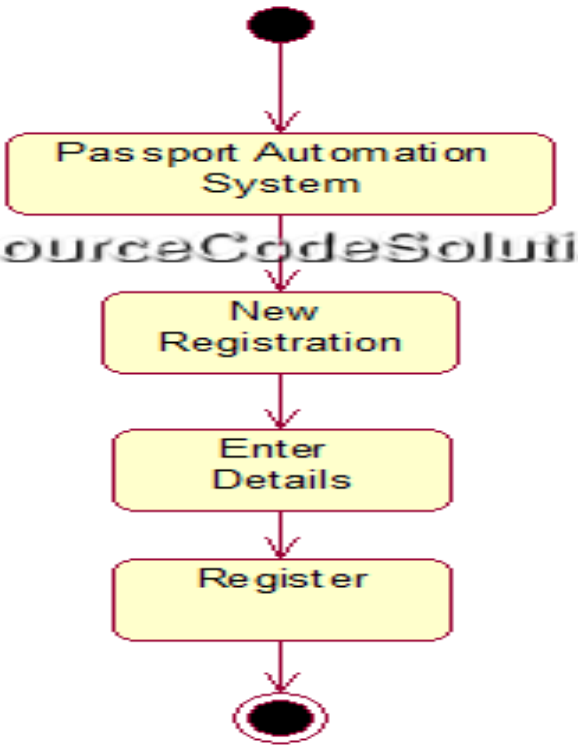
Collaboration Diagram:



State Chart Diagram:



Activity Diagram:



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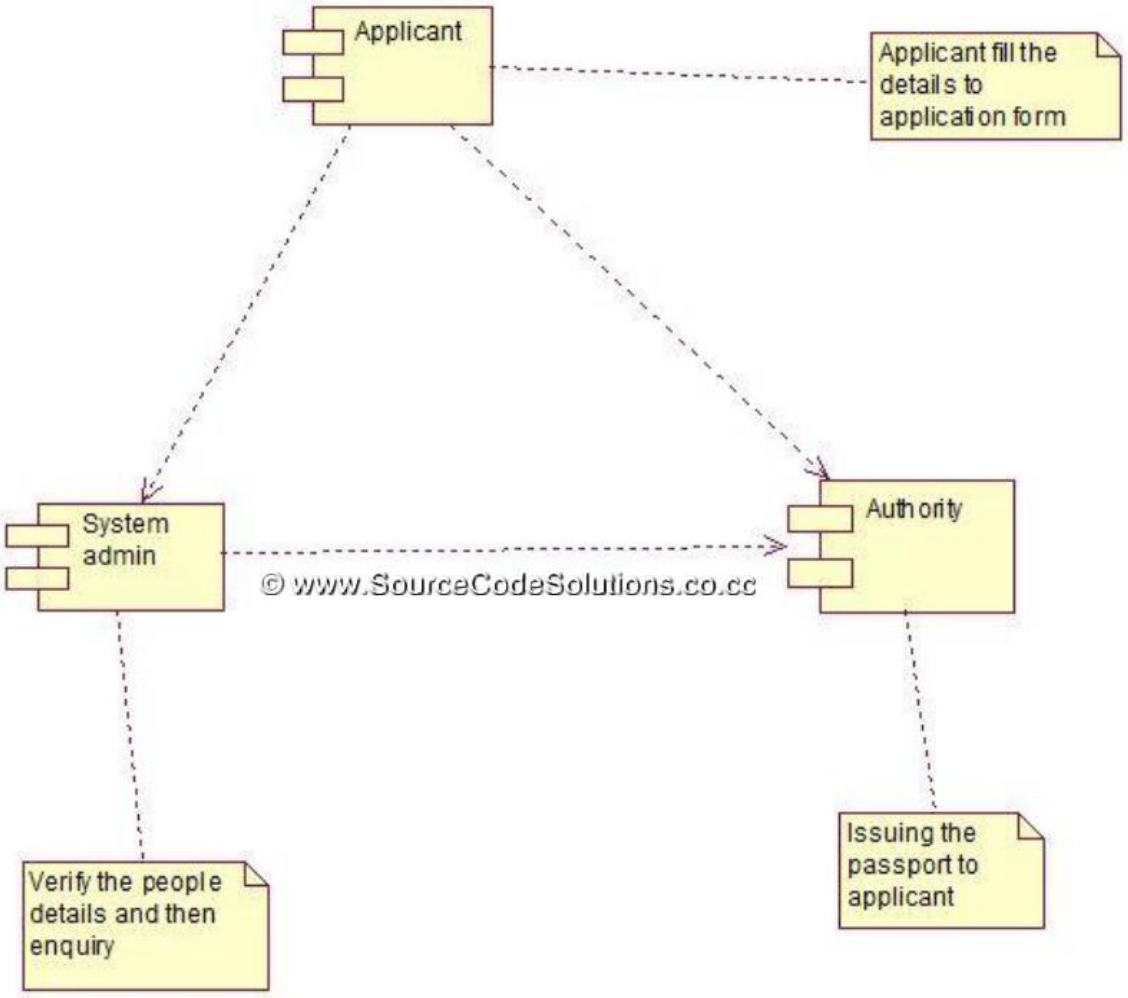
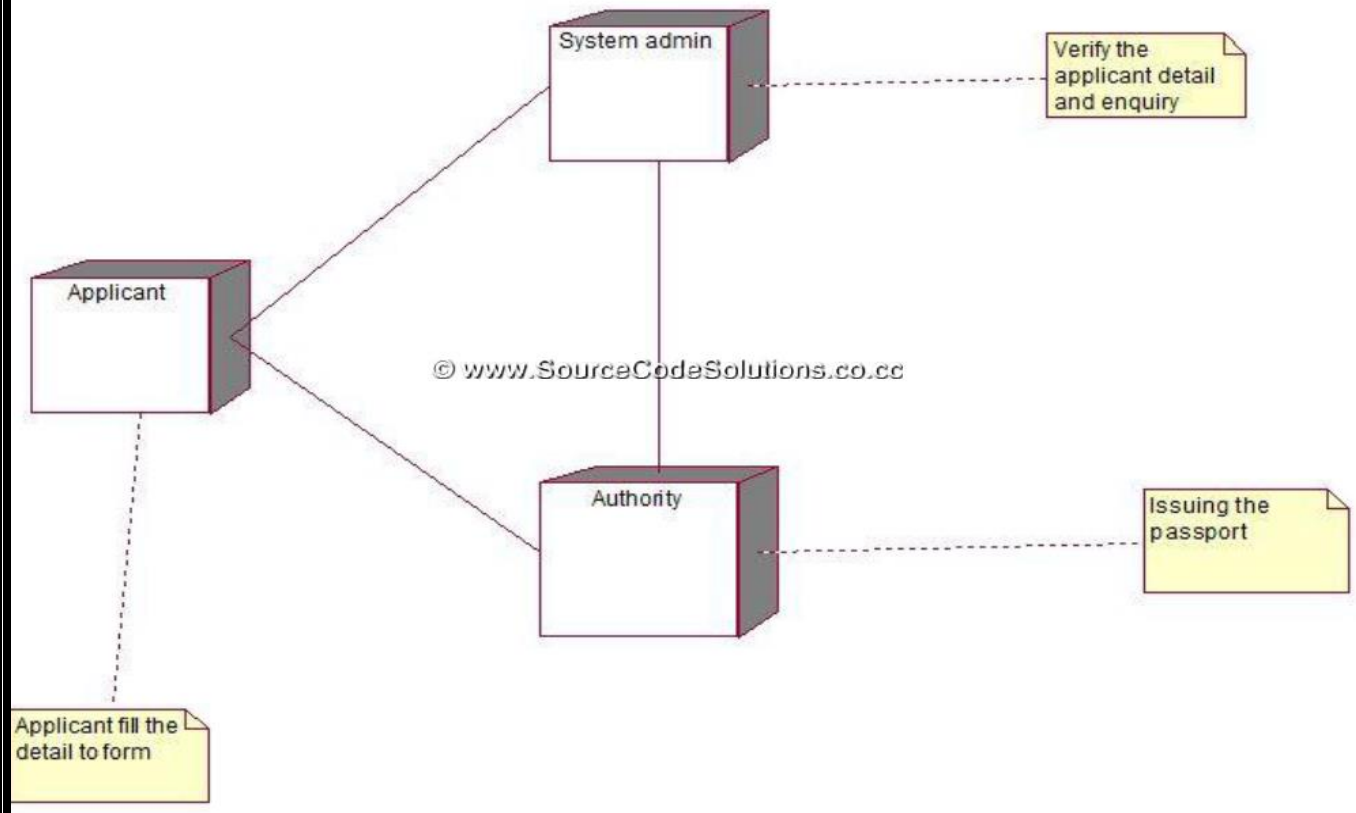
Component Diagram:



Deployment Diagram:



Result:

Ex. No.: 7	UML diagrams for Political Administration System.
Date:	

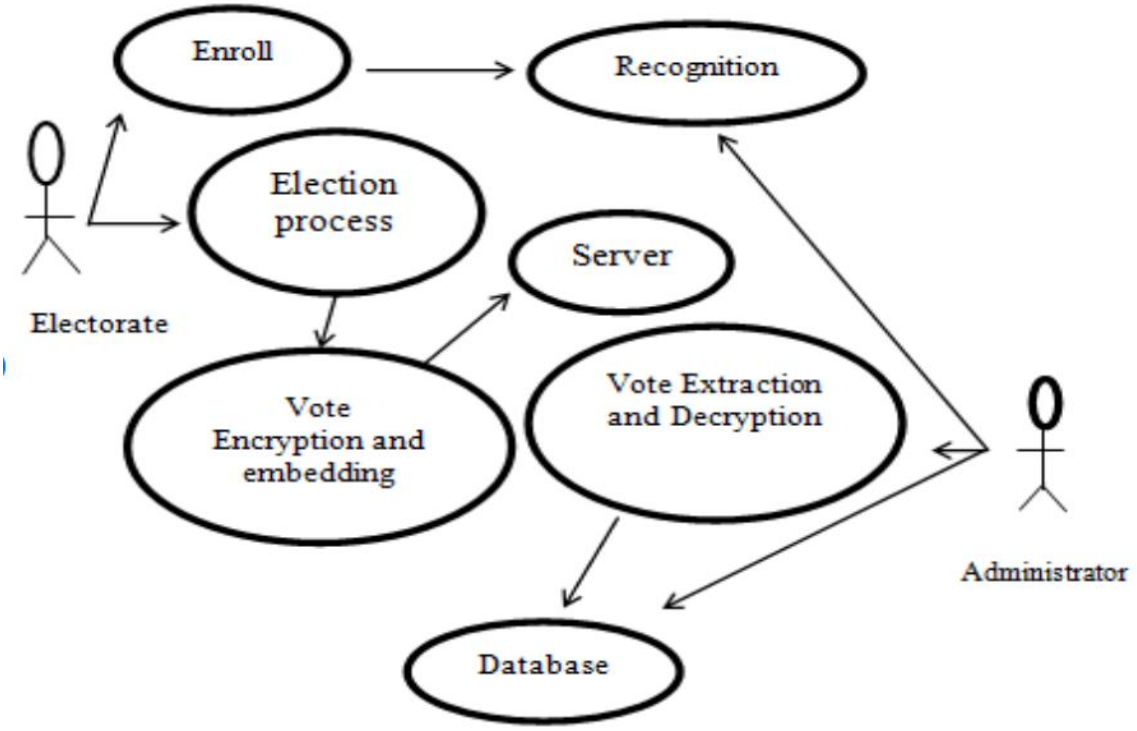
Aim:

To develop a project Political Administration System using Enterprise Architecture Software and to implement the software in Java.

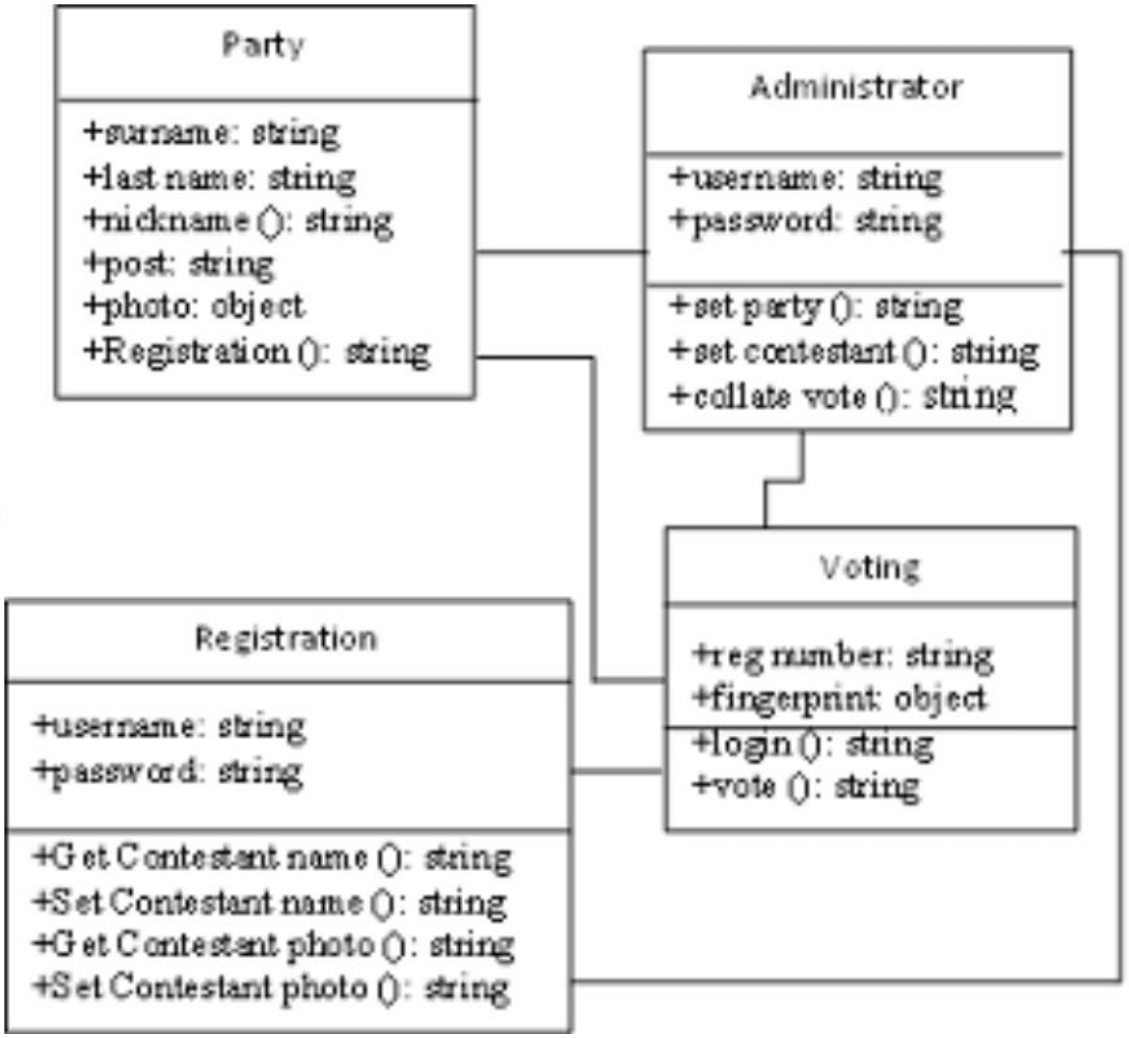
Description:

The study of city management is the study of how politics and administration intersect. While it was conventional in council-manager government at one time to view the governing body in terms of policy making and the administrative staff in terms of implementation, that convention does not pass the empirical test. The city manager sits at the nexus of what is politically acceptable and what is administratively feasible. It is his/her job to facilitate the connection. The key word is facilitate. But others are appropriate as well. The manager "convenes," "builds bridges," and creates "intersections." In all of this, the manager must remain politically sensitive but politically non-aligned. One key to traversing this potential mind field is grounding one's actions in fundamental political values that create legitimacy. A case study is provided to illustrate how one manager's actions reflect this work and the grounding.

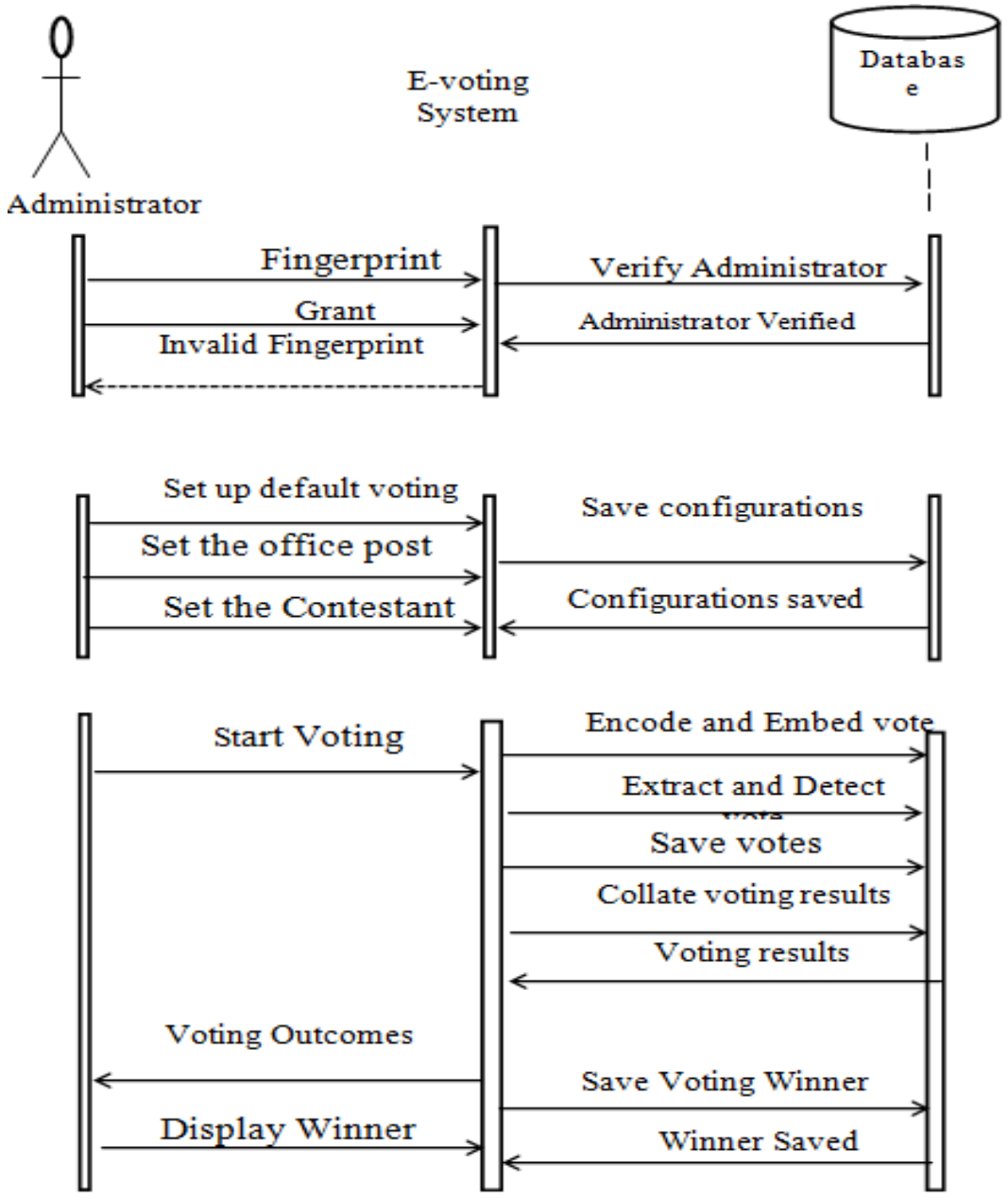
Use case Diagram:



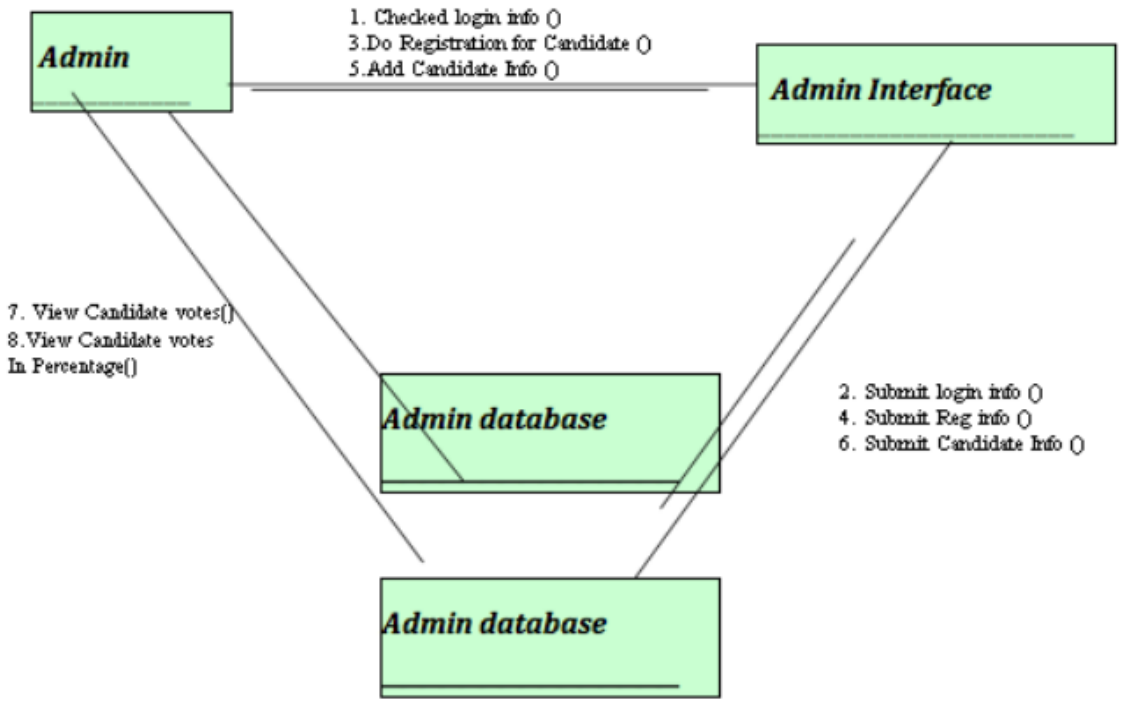
Class Diagram:



Sequence Diagram:

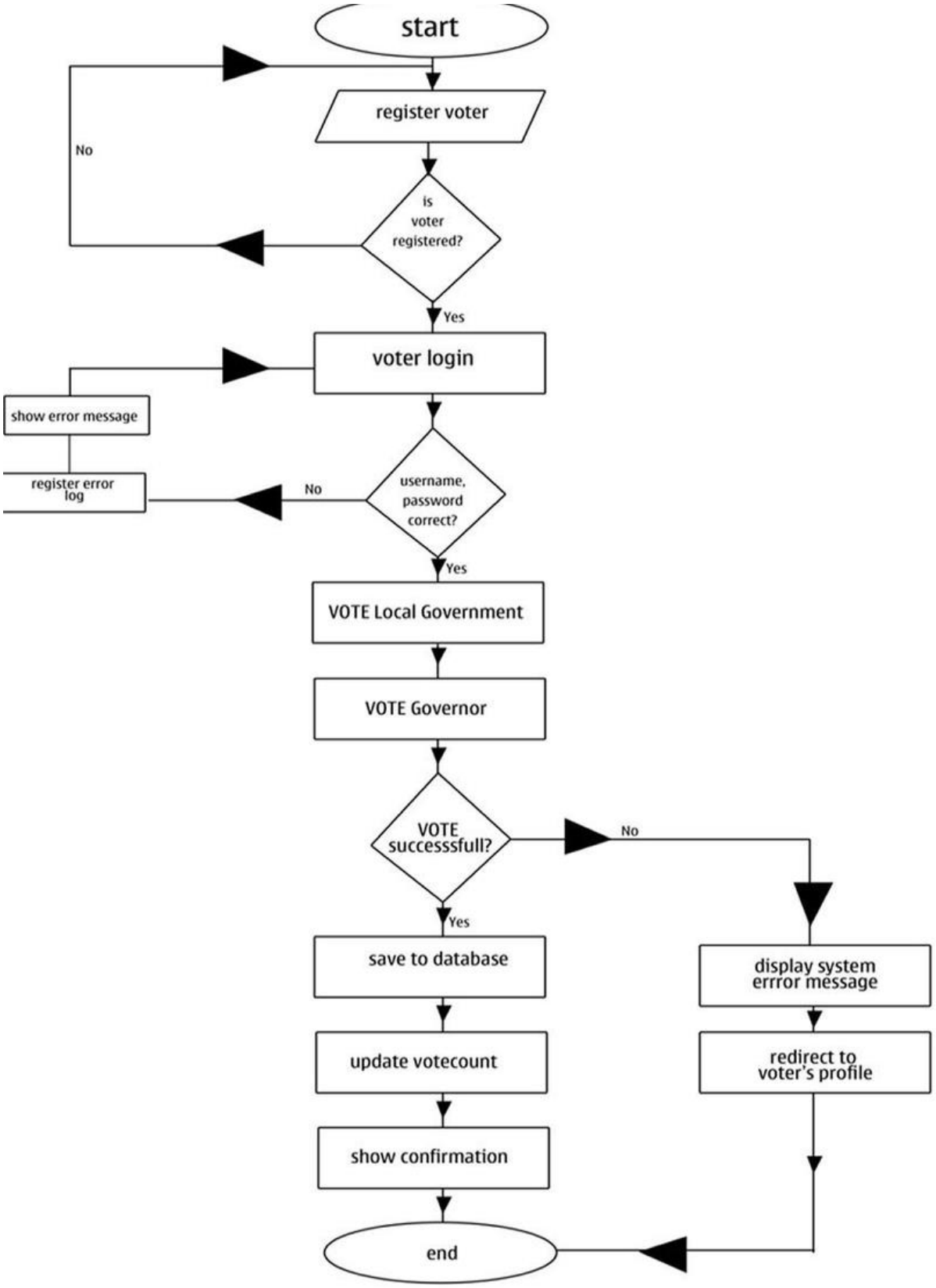


Collaboration Diagram:

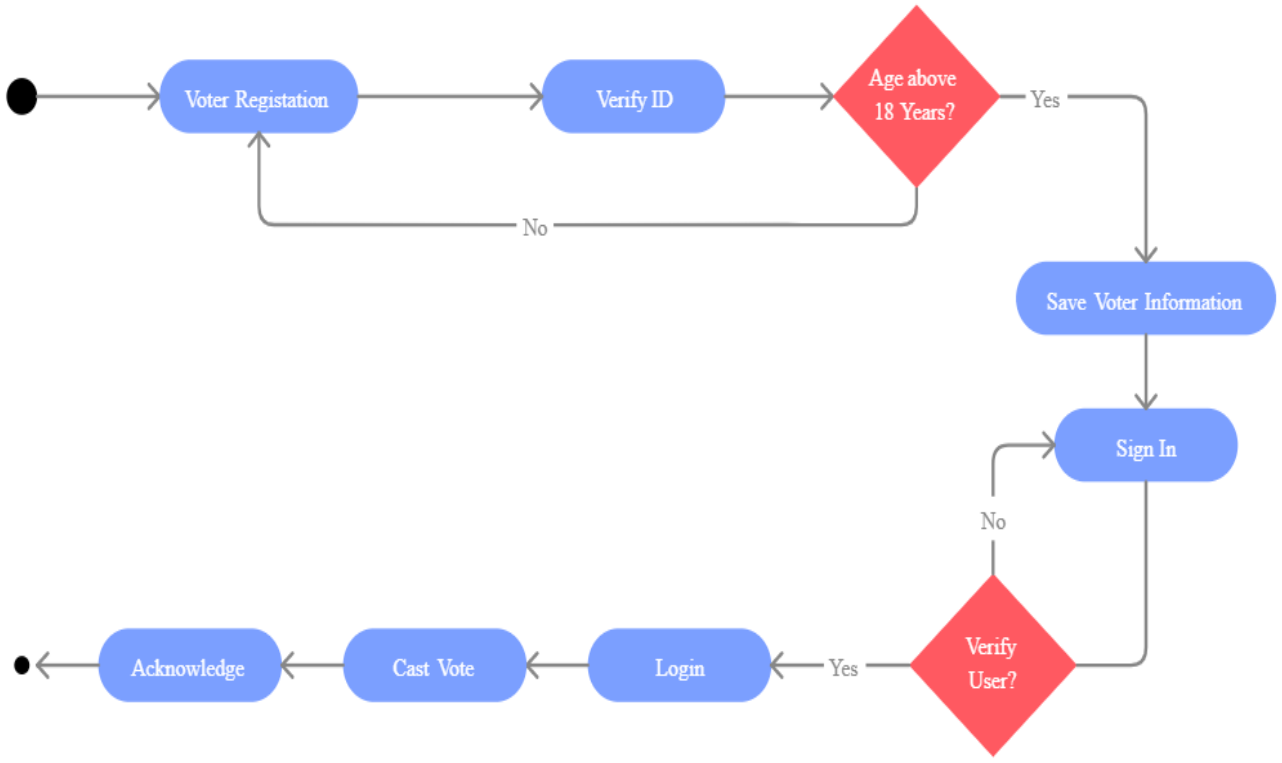


Collaboration Diagram

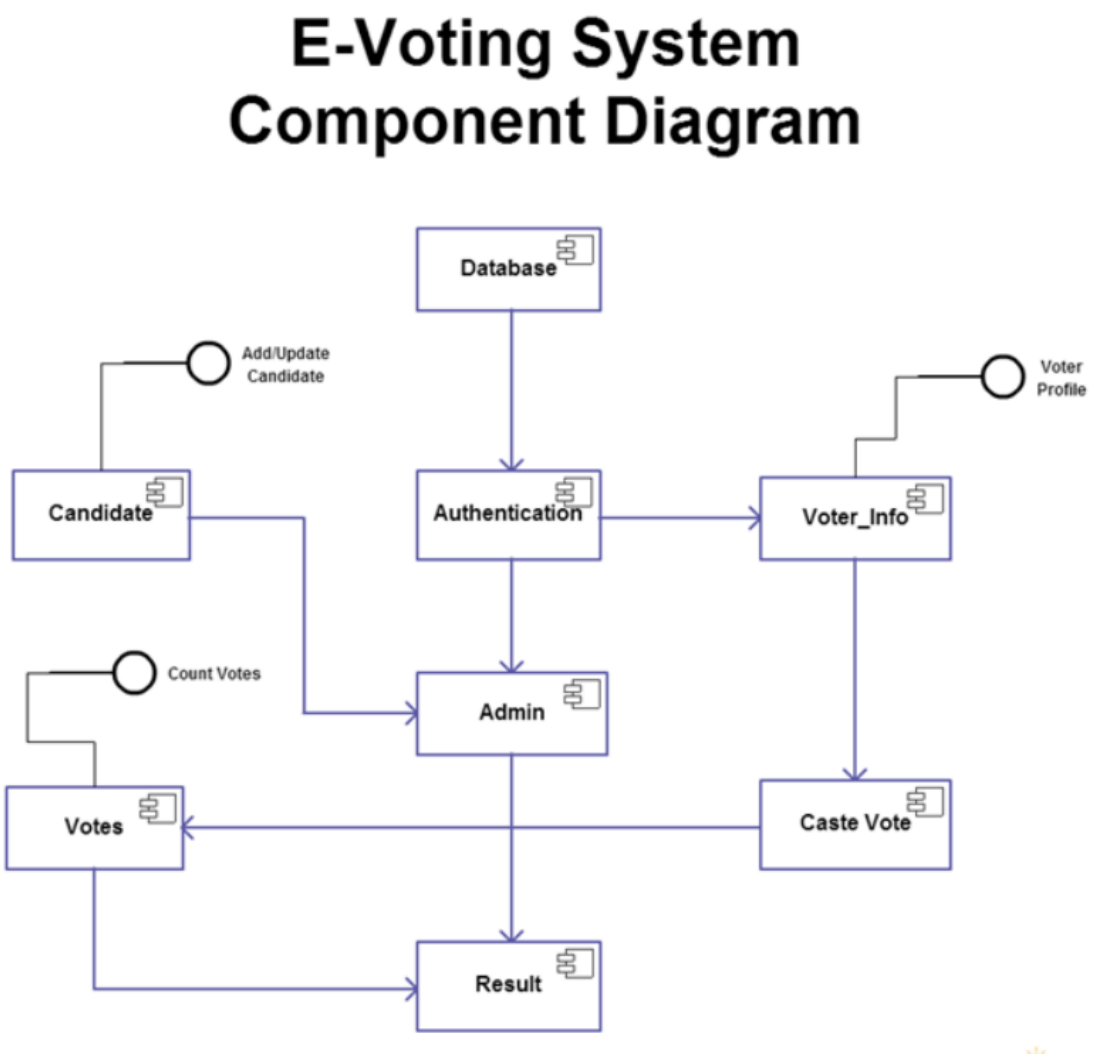
State Chart Diagram:



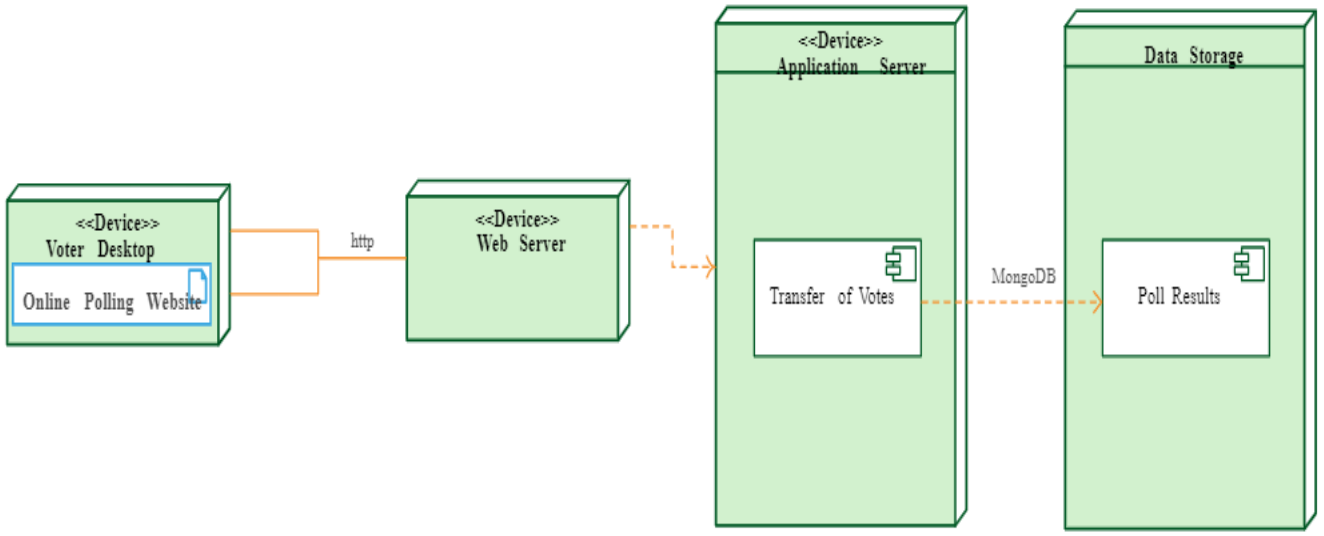
Activity Diagram:



Component Diagram:



Deployment Diagram:



Result: